

Dry Bulk Freight Rate Prediction and Interpretive Analysis: Transformer-Based Hybrid Model with Domain Adaptation

Yuping Luo^{1,2} , Rui Zhang², Yajuan Xiang³, Zhiyuan Liu¹ ,
Wenyang Wang^{2,4,5} , Bohan Wang⁶ , and Jibin Zhou⁴

Transportation Research Record
1–32

© The Author(s) 2026

Article reuse guidelines:

sagepub.com/journals-permissions

DOI: 10.1177/03611981261441288

journals.sagepub.com/home/trr



Abstract

Shipping is vital to global trade but the dry bulk market faces high volatility because of complex factors like economic conditions and geopolitics. Traditional models struggle to capture nonlinear rate fluctuations and offer interpretability. To overcome this, our study integrates shipping, financial, and commodity data. Using the Maximal Information Coefficient (MIC) algorithm, 4,281 time series (over 4.53 million points) were selected to pre-train a general maritime time series model (MTS) based on the Transformer. A fine-tuning dataset of 248 high-MIC series (>0.2) and five binary event indicators (over 0.073 million points) was then used for domain adaptation, creating DA-MTS. To better model short-term changes, a long short-term memory (LSTM) network with Grey Wolf Optimization was applied to DA-MTS residuals, yielding a hybrid DA-MTS+LSTM model. Empirical results show DA-MTS+LSTM achieves high accuracy in forecasting the Baltic Dry Index (MAE=0.16, RMSE=0.21, MAPE=5.26%), the Baltic Capesize Index (MAE=0.21, RMSE=0.27, MAPE=6.67%), and the Baltic Panamax Index (MAE=0.16, RMSE=0.20, MAPE=5.12%). The study also analyzes the short- and long-term impacts of exogenous variables and sudden events. This work enhances prediction accuracy and stability while extending Transformer applications to time series analysis, offering robust theoretical and practical decision-making support.

Keywords

maritime logistics decision support, transformer, freight rate, interpretable AI

Introduction

As a pivotal driver of global economic development, the shipping industry is essential in facilitating international trade and fostering economic growth. Maritime transport accounts for approximately 50% of global trade volume (1), contributing nearly \$380 billion annually to the global economy through freight charges alone. As a crucial segment of the global shipping industry, the dry bulk shipping market intersects with the shipping, finance, and commodities sectors and is profoundly influenced by international economic and geopolitical dynamics. It is widely acknowledged as a high-risk, highly volatile cyclical market (2). Among them, the Baltic Dry Index (BDI) and its sub-indices, Baltic Capesize Index (BCI) and Baltic Panamax Index (BPI), are the most representative freight rates in shipping market, which reflect the overall freight rate levels and serve as an essential barometer of supply and demand dynamics in international trade. In

the current complex and volatile shipping market environment, conducting an in-depth analysis of the impact of factors and unexpected events from the shipping, finance, and commodity markets on freight rates is of great significance. Furthermore, proposing accurate,

¹School of Statistics & Data Science, Southeast University, Nanjing, China

²Collaborative Innovation Center for Transport Studies, Dalian Maritime University, Dalian, China

³School of Economics and Management, Southwest Petroleum University, Chengdu, China

⁴Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian, China

⁵Department of Statistics, University of Missouri, Columbia MO

⁶Economics School, Jilin University, Changchun, China

Corresponding Authors:

Zhiyuan Liu, zhiyuanl@seu.edu.cn

Wenyang Wang, wangwenyang@dlmu.edu.cn

efficient, and market-adaptive methods for freight rate prediction is paramount. Currently, statistical models (SMs), represented by the Autoregressive Integrated Moving Average Model (ARIMA) (3, 4), and machine learning (ML) models, represented by artificial neural networks (ANNs) (5), are extensively utilized in the domain of shipping price forecasting. However, traditional SMs and ML models exhibit certain limitations in capturing the intricate volatility characteristics of freight rates. First, SMs struggle to capture the non-linear characteristics of freight rates. Shipping prices are notably influenced by unexpected factors such as geopolitical risk (GPR) (6), economic crisis, public health events, and economic policy uncertainty (EPU), making it difficult for SMs to capture non-linear fluctuations. Second, the network structures of ML models such as ANN (5), Prophet (7), Temporal Hierarchical Aggregation Tree Structure Network (10), and Graph Convolutional Networks (38) often exhibit instability and are limited in interpretability (7). Moreover, the scope of data utilized in these models is typically constrained to relatively small datasets. For instance, BDI forecasting (8) and China Container Freight Index (CCFI) forecasting (9) predominantly rely on historical data covering limited time periods for both training and testing. Thirdly, shipping prices are usually affected by various financial and commodity market factors, including macroeconomic conditions (11), crude oil price, trend, and seasonal factors (12). However, mainstream SMs and ML models are limited by their structures and cannot incorporate high-dimensional data information, making it challenging to forecast freight rates accurately.

With the rise of Artificial Intelligence (AI) agents such as Llama series (13), the underlying Transformer architecture (14), initially developed as a deep learning (DL) method for natural language processing (NLP), has demonstrated exceptional performance in long-sequence tasks (15). This has sparked significant interest in the academic community to extend its application to time series forecasting, such as Informer (16), TransformEHR (17), Uni-MOF (18) and trRosettaRNA (19). Consequently, this study introduces a novel maritime time series prediction model (MTS) based on Transformer architecture. The model integrates twelve publicly available datasets, encompassing 4,281 time series data (over 4.53 million data points) from shipping, finance, and commodity domains. The integration of these datasets is guided by the Maximum Mutual Information Coefficient algorithm (MIC), with all selected series exhibiting MIC values between 0.1 and 0.2. To further enhance the model's predictive accuracy for dry bulk freight rates, we incorporated five non-public datasets comprising 253 (248 time series from non-public database and five self-constructed

binary sequence indicators) time series (over 0.073 million data points) relevant to shipping indices. This integration specifically targeted variables demonstrating a stronger association, as quantified by an MIC value exceeding 0.2. The MTS model is structurally optimized through transfer learning, and its parameters are adjusted, leading to the development of a domain-adapted MTS for dry bulk market (DA-MTS), which significantly improves the accuracy and convergence speed in BDI, BCI, and BPI prediction. The Transformer architecture, which relies on a self-attention mechanism and is the foundational architecture of MTS and DA-MTS, can be considered an L1-level (20) model. The MTS, based on Transformer architecture, completes parameter optimization on the entire time series datasets, enhancing its DL capabilities across shipping, finance, and commodity domains, and can be viewed as an L2-level model. The DA-MTS, built on MTS, utilizes specific privatized shipping-finance data to perform transfer learning and parameter readjustment, achieving outstanding capabilities in freight rates forecasting and being regarded as an L3-level model.

The proposed MTS and DA-MTS models overcome the limitations of existing forecasting models, demonstrating five significant advantages: (1) Data scale: MTS is developed using public datasets from shipping, finance, and commodity domains, offering comprehensive data coverage. DA-MTS builds on MTS's adaptability across these domains and applies transfer learning to non-public shipping price datasets. By combining large-scale public datasets with small-scale private ones, this dual-layer model overcomes limitations in data diversity and comprehensiveness. (2) Accuracy: Unlike models based on feature combinations and weight adjustments, MTS and DA-MTS use the self-attention mechanism to fully capture key dependencies in time series. This approach effectively addresses long-, medium-, and short-term impacts, identifies cyclical changes and seasonal fluctuations in dry bulk freight rates, and ensures accurate and stable forecasting results. (3) Interpretability: DA-MTS enhances model transparency by outputting feature importance, making the decision-making process more transparent. Existing neural network models have complex structures and intricate internal decision-making processes, resulting in low interpretability. (4) Extended Application: As a cross-domain forecasting framework, MTS can not only be used for transfer learning to train the DA-MTS model but also for other forecasting models in shipping, finance, and commodity domains or across these fields, which demonstrates its powerful generalization capability. (5) Efficiency: MTS learns general feature representations from large-scale datasets, enabling it to adapt to new tasks quickly. This allows for zero-shot or few-shot learning, thereby avoiding the need

for model reconstruction and significantly enhancing the model's efficiency.

Although DA-MTS excels in forecasting shipping indices, its training on large-scale data emphasizes long-term trends, limiting its ability to capture sudden short-term fluctuations. During events like the 2008 financial crisis, Brexit, COVID-19, the Russia–Ukraine conflict, and the Red Sea crisis, its performance in predicting short-term shipping price changes still needs improvement. Figure 3 illustrates the residual fluctuation characteristics of DA-MTS in forecasting three major dry bulk freight rates, confirming DA-MTS's insufficiency in short-term trend forecasting. Considering this, this paper further introduces a long short-term memory (LSTM) network and designs a unique Grey Wolf Optimization (GWO) algorithm to estimate the parameters of LSTM model. This approach allows LSTM to accurately capture short-term fluctuations in freight rates within the residuals of DA-MTS's predictions. As a swarm intelligence algorithm, GWO provides accurate estimation results and enhances the convergence speed of parameter estimation through parallel computation using multi-core central processing units (CPUs) or graphics processing units (GPUs).

In a word, this paper proposes a hybrid model (DA-MTS + LSTM) for dry bulk freight rate prediction (BDI, BPI, and BCI). With regard to data, this study collects information from public databases in shipping, finance, and commodity domains to construct the general model (MTS). It introduces shipping-finance commercial database information related to freight rates and builds a private database that includes 248 exogenous variables significantly related to freight rate indices across the three markets and five binary variables representing sudden events to construct the domain-adapted model (DA-MTS). Furthermore, to enhance the interpretability of the combined model, this paper utilizes a stepwise elimination method of exogenous variables and a category-wise elimination method to adjust the exogenous variables in the DA-MTS's and LSTM's prediction process. The importance of different factors for long- and short-term trend prediction in freight rates is evaluated by observing changes in error metrics (Mean Absolute Error, MAE) of the forecasting results. Figure 1 shows the research framework of this paper. The structure of this paper is as follows: the next section provides a review and summary of shipping price prediction methods, the factors influencing freight rates, and relevant literature on Transformer applications; details about the construction of MTS, DA-MTS, and LSTM models employed in this paper are then provided; a section then presents the data description and empirical analysis results of this paper; and the final section concludes the research and proposes future research prospects.

Literature Review

Review of the Literature on Freight Rate Prediction

Accurately forecasting freight rates and cycles is essential for business decision-makers (21). With regard to models, the academic community has primarily used Vector Autoregressive (VAR), ANN (5), Prophet (8) and link prediction method (22) for freight rate prediction. Concerning research subjects, the academic community has primarily focused on Shanghai Containerized Freight Indices (SCFI) (3), container rates (7), BDI (8), BPI (23), other freight rates (24), and charter rates (25). Academics increasingly acknowledge that a single model cannot fully capture key features of complex sequences. Integrating SMs and ML models to decompose time series for more accurate forecasting has become a mainstream approach. This study introduces an ensemble strategy combining SMs (MIC), ML method (GWO), DL model (LSTM), and Transformer models (MTS, DA-MTS), resulting in the DA-MTS + LSTM algorithm, which significantly improves the accuracy, stability, and efficiency of freight rate prediction. The research methods for freight rate prediction are summarized in Table 1.

Study of Factors Influencing Freight Rates

The dry bulk shipping market is significantly influenced by the supply and demand for transportation in the bulk commodity market, fuel cost prices, and the macroeconomic environment. Under the impact of market factors and unexpected events, the fluctuations in dry bulk freight rates are exceptionally complex. Market factors such as oil prices (26), iron ore and agricultural product prices (27), GPR (28, 29), China economic policy uncertainty (CEPU), and global economic policy uncertainty (GEPU) all cause significant fluctuations in freight rates. At the same time, the characteristics of charterers and shipowners are also important microeconomic determinants of freight rates (30). From the perspective of unexpected events, the signing of the Paris Agreement, Qatar's withdrawal from the Organization of the Petroleum Exporting Countries (OPEC) in 2019, financial crises (8), and COVID-19 (31) all directly or indirectly affected freight rate indices. Previous studies have examined the impact of individual factors on shipping prices, focusing on market factors and unexpected events, but two gaps remain. First, the analysis has been limited to a subset of factors, lacking a systematic comparison of their effects on freight rates. Second, research on the impact of unexpected events, such as holidays, geopolitical events, and economic crises, is insufficient. Therefore, this paper systematically analyzes the effects of 248 market factor variables and five binary sequence variables for unexpected events, significantly related to dry bulk freight rates, with the MIC method and data

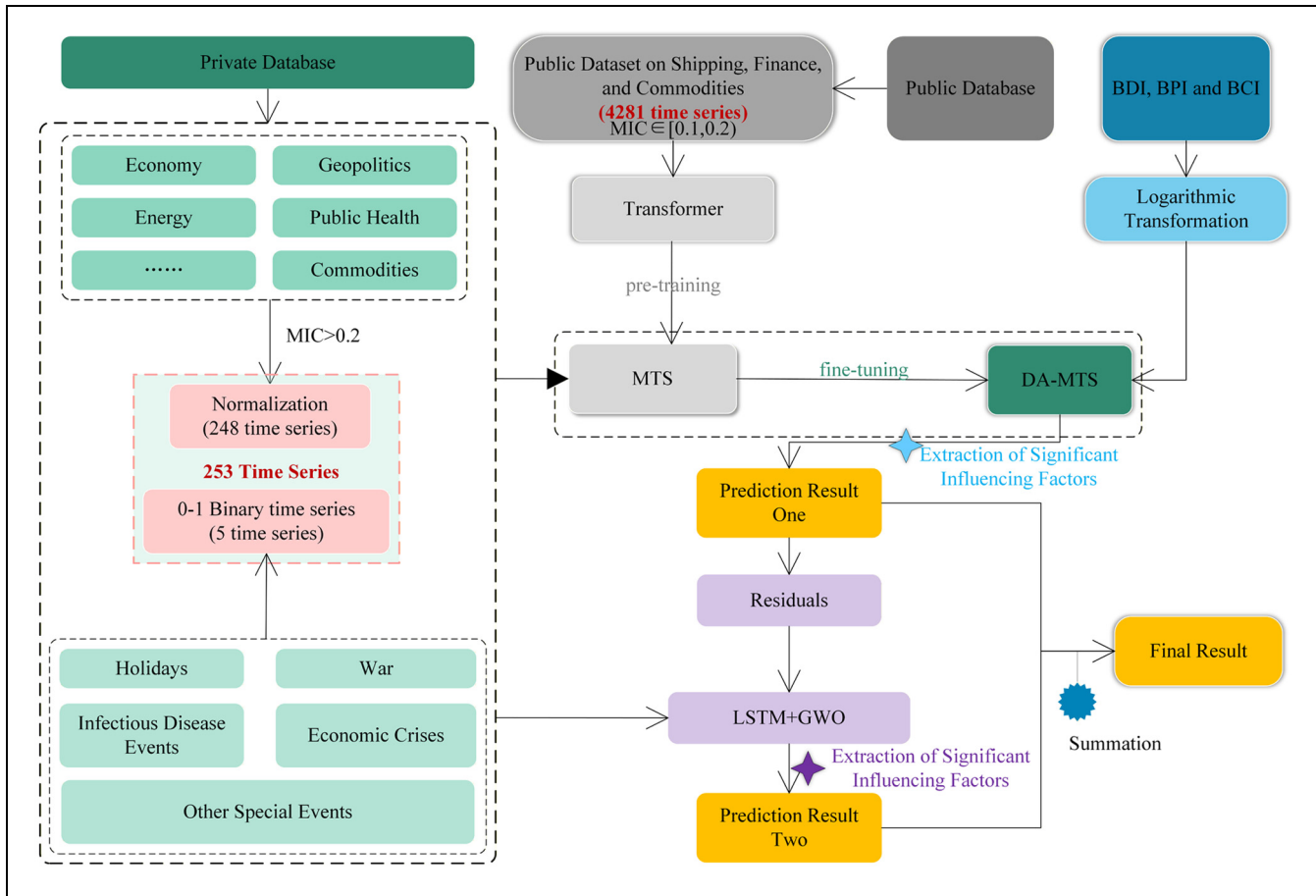


Figure 1. Research framework of dry bulk freight rate indices.

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; MTS = maritime time series prediction model; DA-MTS = domain-adapted maritime time series prediction model for the dry bulk market; LSTM+GWO = a hybrid model integrating a long short-term memory network with Grey Wolf Optimization; MIC = maximum mutual information coefficient.

Table 1. Summary of Freight Rates Forecast Research Methods

Model	Category
ARIMA (3, 4)	SMs
ANN (5)	ML
Prophet (7, 8)	
This paper: DA-MTS+LSTM	Integration of SMs, DL, ML, and Transformer

Note: ARIMA = autoregressive integrated moving average; SMs = statistical models; ANN = artificial neural network; ML = machine learning; DA-MTS = domain-adapted maritime time series prediction model for the dry bulk market; LSTM = long short-term memory; DL = deep learning.

from commercial and private databases. The study of factors influencing freight rates is shown in Table 2.

Research on the Applications of Transformer

Recent research findings have confirmed the outstanding performance advantages of Transformer architecture in

Table 2. Summary of Literature on Factors Influencing Freight Rates

Influencing factor	Category
Bulk commodities (27)	Market factors
Geopolitical risk (29)	
Characteristics of charterers and shipowners (30)	
Multiple types of unexpected events (8)	Unexpected events
COVID-19 (31)	
This paper: Covering all factors mentioned in previous literature	Comprehensive

sequence modeling (14, 32, 33). For example, in the transportation field, TrafficBERT is used for traffic flow prediction and significantly outperforms traditional SMs and DL models with regard to accuracy (34); in the materials field, TransPolymer is used for polymer performance prediction, and compared with the baseline

Table 3. Transformer Applications

Field	Objective	Model name
Transportation	Traffic flow forecast	TrafficBERT (34)
Materials	Polymer performance prediction	TransPolymer (35)
Climate and earth sciences field	three-dimensional (3D) upper-ocean temperature anomalies and wind stress anomalies	3D-Geoformer (36)
Power	Wind power forecast	BERT4ST (37)
Biology	Toxicity forecast	AI models based on the Transformer architecture (41)
Water resources management field	Water flow forecast	MVMD-Transformer (43)
Financial	Financial market volatility	Hybrid neural network model (44)
Aviation	Aviation fault event forecast	STL-transformer-ARIMA (45)
Energy, economy, etc.		MAEformer (46)
		Robformer (47)
Maritime	Ship prices	SPPformer (50)
This paper: Shipping-finance	Freight rates forecast	DA-MTS
This paper: Advantages	Integrating statistical models, machine learning method, deep learning model, and Transformer, it is currently the most stable, interpretable, efficient, accurate, flexible, and adaptable freight rate prediction model in the shipping-finance field. A novel cross-domain prediction framework that can quickly adapt to new tasks based on dynamic adaptability, effectively solving cross-domain prediction problems, and providing accurate and reliable prediction results.	

Note: DA-MTS = domain-adapted maritime time series prediction model for the dry bulk market.

models like random forests (RF), TransPolymer's root mean square error (RMSE) decreases by an average of 7.7% and R^2 increases by an average of 11% (35); in the climate and earth sciences fields, 3D-Geoformer is used for the three-dimensional forecast of upper ocean temperature anomalies and wind stress anomalies (36); in the power field, the BERT4ST model (37) is used for long-term wind power forecast based on the North Sea dataset, and compared with the STGCN (38) and MTGNN models (39) MAE decreases by 31.52% and 7.35%, respectively; in the chemical engineering field, the CEGPT-PF-M model is utilized for predicting the China methanol price, achieving an MAE of only 0.0859 and a RMSE of merely 0.1072 (40); in the field of biology, AI models based on Transformer architecture are used for toxicity prediction (41); in water resources management, Transformer outperforms convolutional neural networks (CNNs) and feedforward neural networks (FNNs) in groundwater level prediction (42); the Nash–Sutcliffe Efficiency of MVMD-Transformer reaches 0.85, significantly outperforming Transformer and LSTM (43); in the financial field, hybrid neural network models based on Transformer architecture are used for volatility forecasting in financial markets (44); in the aviation field, the STL-transformer-ARIMA model is used for aviation fault event prediction, and compared with temporal convolutional network (TCN) and LSTM, MAE decreases by an average of 10.4% and mean absolute percentage error (MAPE) decreases by 9.7% (45). Additionally,

MAEformer (46) and Robformer (47) perform excellently in multiple fields such as energy, economy, transportation, weather, and disease prediction, with Robformer achieving performance improvements of 17% and 10% compared with Autoformer (48) and FEDformer (49), respectively; in the shipping domain, when predicting new ship prices, SPPformer achieves an average MAE and RMSE of 0.2432 and 0.2814, respectively (50). This paper develops MTS and DA-MTS models based on Transformer architecture, integrating publicly available time series datasets from shipping, finance, and bulk commodity sectors. Transformer applications in various fields are shown in Table 3.

Model Framework

Correlation Analysis

This paper uses the MIC algorithm (51) to analyze the correlation between exogenous variables, unexpected events, and the dependent variable of the freight rate indices to demonstrate the rationality of the explanatory variables selected in this paper. The MIC algorithm is based on mutual information that measures the strength of the association between two variables, defined as follows: Suppose X and Y are two random vectors in the dataset, where $X = \{x_1, \dots, x_n\}$ and $Y = \{y_1, \dots, y_n\}$, n is the number of the samples. The mutual information between X and Y is given by:

$$I(X; Y) = \int_Y \int_X f(X, Y) \log \frac{f(X, Y)}{f(X)f(Y)} dX dY, \quad (1)$$

where $f(X, Y)$ is the joint probability density function, and $f(X)$ and $f(Y)$ represent the marginal probability densities of X and Y , respectively. The MIC algorithm can maximally identify the linear and nonlinear associations between the candidate input variable X and the output variable Y ; the calculation formula is:

$$MIC(X; Y) = \max_{m \times n < B} \frac{I(X; Y)}{\log \min(m, n)}, \quad (2)$$

where B is the maximum upper limit for constructing an $m \times n$ grid, which is a function related to the sample size n . The stronger the correlation between X and Y , the closer the value of $MIC(X; Y)$ approaches 1; conversely, if X and Y are independent, $MIC(X; Y)$ approaches 0.

MTS and DA-MTS Model Structures

MTS leverages the self-attention mechanism to globally analyze time series data, capturing interdependencies between time points and enhancing data representation with local positional encoding. The model, comprising approximately 11 million parameters, was fine-tuned on a dataset of 4,281 time series (over 4.53 million data points). This optimization was performed through a forward propagation process across the encoder–decoder architecture. Each encoder and decoder layer incorporates multi-head attention mechanisms, layer normalization, residual connections, and FNN, ultimately mapping the decoder's output to the prediction window dimension with a linear layer. Built on the Transformer architecture, MTS effectively handles time series data with varying frequencies and characteristics, showing strong adaptability to diverse input sizes and forecast ranges. It is designed to minimize forecast errors, adhering to the principles of training large Transformer models on extensive datasets. Specifically optimized for time series data in shipping, finance, and commodities, MTS serves as the core model of this study, demonstrating excellent forecasting performance in shipping finance. With a diverse training foundation, it features zero-shot and few-shot inference capabilities, robust generalization, and the ability to deliver accurate predictions within error margins, even when applied to unfamiliar datasets. To meet the demand for specialization in freight rate prediction, we introduce non-public commercial databases and private shipping freight rate-related data to fine-tune MTS model, and after multiple rounds of adjusting the parameters of the encoder, FNN layers, and linear layers, the DA-MTS model is trained to capture the fluctuation trends of freight rates accurately and transforms them into high-accuracy prediction results. The structures of MTS and

DA-MTS models are shown in Figure 2. The detailed hyperparameters, training configurations, and parameter counts for both the MTS and DA-MTS models are provided in Table 4. Although DA-MTS demonstrates strong performance in capturing long-term trends and nonlinear patterns in freight rate indices, its architecture, based on the Transformer model, is inherently designed to focus on global dependencies and multi-scale trends. This makes it less sensitive to short-term high-frequency fluctuations, as shown in Figures 3 and 8. To address this limitation, we introduce an LSTM model to capture the short-term fluctuations present in the residuals of DA-MTS predictions. LSTM is particularly well suited for this task because of its gated structure, which includes input, forget, and output gates. These gates allow LSTM to selectively retain relevant short-term information while discarding noise, making it highly effective for modeling residual volatility. Compared with other residual modeling strategies, such as ARIMA, GRU, and Prophet, LSTM offers the following advantages: (1) Handling nonlinear dependencies: Unlike ARIMA, which is limited to linear relationships, LSTM can model complex nonlinear patterns in residuals; (2) memory retention: LSTM's ability to maintain short-term memory over multiple time steps makes it more effective than other ML models in capturing intricate short-term dependencies; (3) robustness to noise: The gating mechanism of LSTM allows it to filter out irrelevant noise, which is particularly important for volatile residual data. As shown in Table 5, LSTM outperforms ARIMA, GRU, Recurrent Neural Network (RNN), and Prophet with regard to MAE, RMSE, and MAPE when modeling residuals of DA-MTS predictions.

LSTM Time Series Prediction Model

LSTM Model. DA-MTS has a limited ability to capture short-term sudden fluctuations information. Therefore, this paper introduces the LSTM model to accurately capture short-term fluctuations in freight rates within the residuals of DA-MTS's predictions. LSTM is a particular RNN that can maintain short-term memory over multiple steps and performs outstandingly when dealing with long-term serial data. LSTM structure includes input gates, output gates, forget gates, and memory cells. It filters irrelevant information and selectively retains important information through these four gate structures. To enhance the performance of the LSTM model, this paper designs a unique GWO algorithm to optimize LSTM hyperparameters, improving the model's sensitivity to short-term fluctuations in freight rates. The detailed hyperparameters, training configurations, and parameter counts for the LSTM model are provided in Table 6.

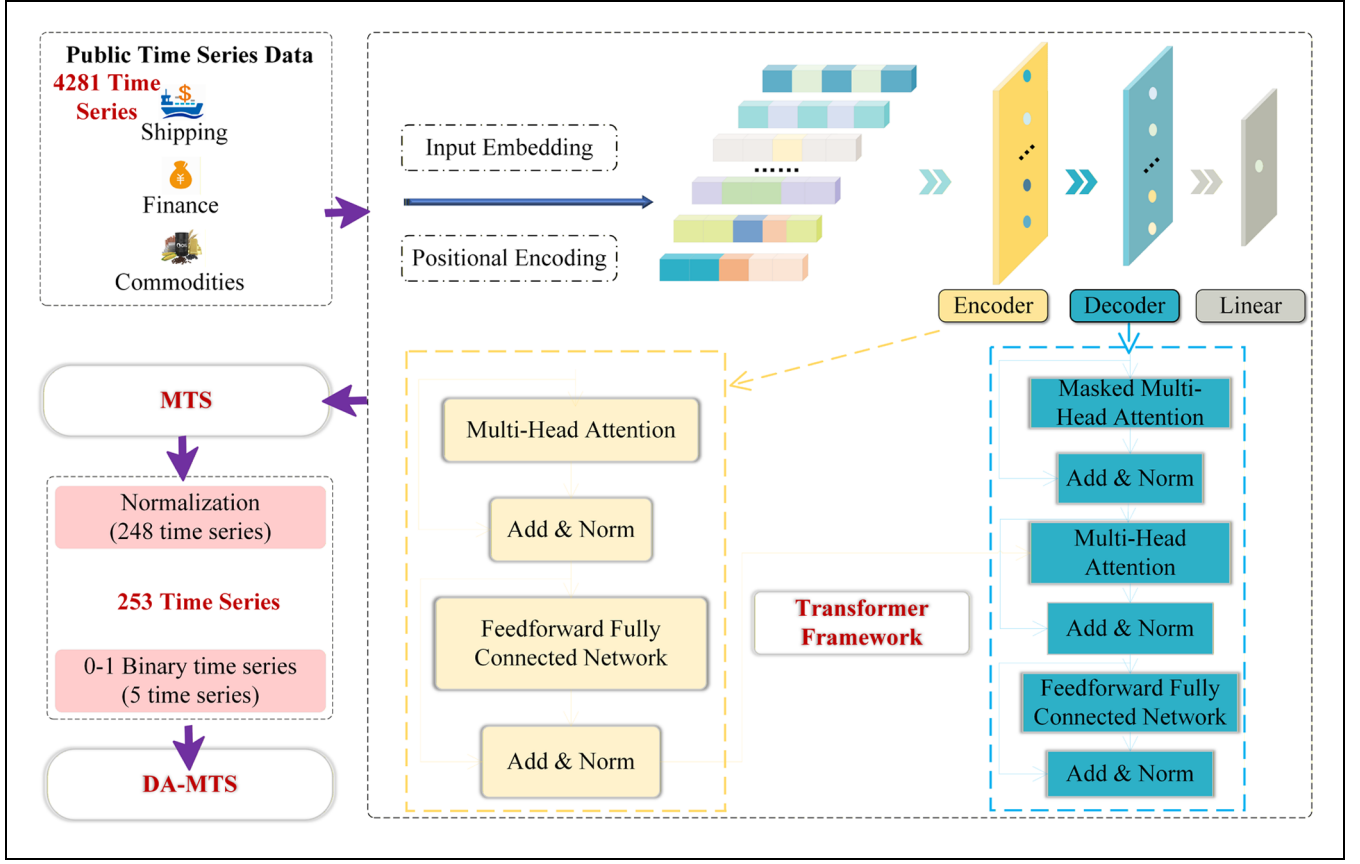


Figure 2. The framework of MTS and DA-MTS.

Note: MTS = maritime time series prediction model; DA-MTS = domain-adapted maritime time series prediction model.

Table 4. MTS and DA-MTS Hyperparameters

Hyperparameter name	Parameter value
learning rate	1e-4
maximum training epochs of MTS model	100
maximum training epochs of DA-MTS model	150
optimizer	Adam
loss function	Mean squared error
num_layers	12
nhead	12
dmodel	384
batch size	16
dropout	0.2
training device	NVIDIA GeForce RTX 4070 (12GB)
training time for MTS	36 hours
dataset size (public)	4281 time series
dataset size (non-public)	253 time series
training time for DA-MTS	3 hours
parameter size	11 M

Note: MTS = maritime time series prediction model; DA-MTS = domain-adapted maritime time series prediction model.

The forget gate f_t determines which information to retain and forget from the memory cell:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (3)$$

the input gate i_t controls the flow of information into the memory cell:

$$\begin{aligned} i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\ \hat{C}_t &= \tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \\ C_t &= f_t \cdot C_{t-1} + i_t * \hat{C}_t \end{aligned} \quad (4)$$

the output gate O_t is responsible for outputting information and determining the hidden state h_t :

$$O_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o), \quad (5)$$

$$h_t = O_t \cdot \tanh(C_t), \quad (6)$$

in the equations, C_t is the cell state; $\sigma(\cdot)$ is Sigmoid activation function. W is weight matrix; b is bias vector; h_t is the hidden state at time t ; $\tanh$ represents the activation function used to compute the new cell state $\hat{C}_t$.

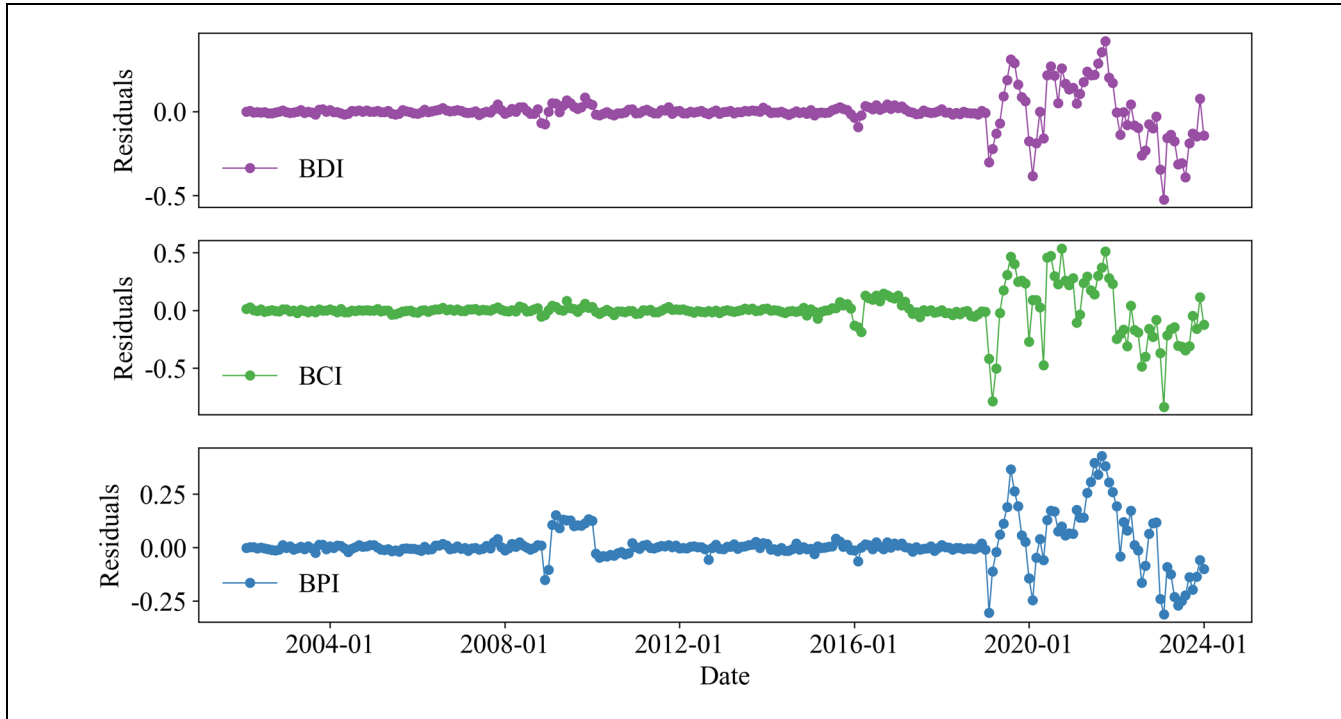


Figure 3. The residual volatility analysis of DA-MTS prediction results.

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; DA-MTS = domain-adapted maritime time series prediction model.

Table 5. Comparison of Residual Modeling Strategies

Models	BDI			BCI			BPI		
	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %
ARIMA	0.090	0.14	4.20	0.14	0.19	4.50	0.088	0.13	3.22
RNN	0.078	0.12	3.10	0.16	0.16	4.10	0.075	0.11	3.07
Prophet	0.085	0.13	3.70	0.13	0.18	4.20	0.082	0.12	3.51
LSTM	0.065	0.097	2.50	0.10	0.14	3.90	0.063	0.093	2.34
GRU	0.070	0.10	2.80	0.12	0.15	4.20	0.068	0.098	2.63

Note: BDI, baltic dry index; BCI, baltic Capesize index; BPI, baltic Panamax Index; MAE, Mean Absolute Error; RMSE, Root Mean Square Error; MAPE, Mean Absolute Percentage Error; ARIMA, Autoregressive Integrated Moving Average; RNN, Recurrent Neural Network; LSTM, Long Short-Term Memory; GRU, Gated Recurrent Unit.

Table 6. Long Short-Term Memory Hyperparameters

Hyperparameter name	Parameter value
learning rate	1e-4
number of hidden units	32
maximum training epochs	500
optimizer	Adam
dropout	0.3
batch size	16
num_layers	4
parameter size	0.06 M

GWO Algorithm. The hyperparameter combination directly determines the forecasting performance of LSTM. To search for the optimal parameter combination, this paper uses the GWO algorithm to optimize the LSTM hyperparameters. GWO is an optimization algorithm based on wolves' social behavior patterns, which simulates wolves' predatory behavior to optimize problem solutions. Combined with the definition of the GWO algorithm, the optimization steps are shown in Algorithm 1.

Algorithm 1. (Grey Wolf Optimization Algorithm)

Input: Initialize the grey wolf population a ; max_iter ; $\vec{A}$ and $\vec{C}$.
 Step 1: Calculate the fitness of each grey wolf individual and save the top three wolves with the best fitness as α , β and δ .
 for $t = 1, 2, \dots, max_iter$ do
 Step 2: Update the positions of the current grey wolf individuals:
 for each $\vec{X}$ in population do
 $\vec{D}_\alpha = |\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}|$
 $\vec{D}_\beta = |\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}|$
 $\vec{D}_\delta = |\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}|$
 $\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \cdot (\vec{D}_\alpha)$
 $\vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \cdot (\vec{D}_\beta)$
 $\vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \cdot (\vec{D}_\delta)$
 $\vec{X}(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3}$
 Step 3: update a , $\vec{A}$ and $\vec{C}$.
 Step 4: Update the fitness values and positions $\vec{X}_\alpha$, $\vec{X}_\beta$
 and $\vec{X}_\delta$ of α , β and δ ;
 end for
 end for
 Output: $\vec{X}_\alpha$, $\vec{X}_\beta$ and $\vec{X}_\delta$, that is, output the optimal parameter combination.

DA-MTS+LSTM Hybrid Model. To accurately capture freight rate fluctuations and improve forecast accuracy, this paper uses the DA-MTS+LSTM model to forecast the dry bulk market freight rates. The overall architecture of the DA-MTS+LSTM model is presented in Figure 1. The specific steps are shown in Algorithm 2. The mathematical formulation of the DA-MTS+LSTM model is as follows: Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the multivariate time series input, where n is the number of time steps and d is the number of features. The MTS model processes the input sequence to generate a predicted output:

$$\hat{\mathbf{Y}}_{DA-MTS} = DA - MTS(\mathbf{X}), \quad (7)$$

where $\hat{\mathbf{Y}}_{DA-MTS} \in \mathbb{R}^{n \times 1}$ represents the predicted freight rates. The residuals are calculated as the difference between the ground truth and the DA-MTS prediction:

$$\mathbf{R} = \mathbf{Y} - \hat{\mathbf{Y}}_{DA-MTS}, \quad (8)$$

where $\mathbf{Y} \in \mathbb{R}^{n \times 1}$ is the ground truth freight rates and $\mathbf{R} \in \mathbb{R}^{n \times 1}$ is the residual series. The residual series $\mathbf{R}$ is fed into the LSTM framework to capture additional patterns.

$$\hat{\mathbf{R}} = \text{LSTM}(\mathbf{R}), \quad (9)$$

Algorithm 2. (Freight Rate Prediction Algorithm)

Input: y_{BDI} , y_{BCI} and y_{BPI} .
 Step 1: DA-MTS forecast to obtain $\hat{y}_{BDI}$, $\hat{y}_{BCI}$ and $\hat{y}_{BPI}$.
 Step 2: Calculate residuals $\epsilon_{BDI} = y_{BDI} - \hat{y}_{BDI}$, $\epsilon_{BCI} = y_{BCI} - \hat{y}_{BCI}$ and $\epsilon_{BPI} = y_{BPI} - \hat{y}_{BPI}$.
 Step 3: LSTM forecasts the residuals to obtain $\hat{\epsilon}_{BDI}$, $\hat{\epsilon}_{BCI}$ and $\hat{\epsilon}_{BPI}$.
 Step 4: Calculate the final forecast results $\tilde{y}_{BDI} = \hat{y}_{BDI} + \hat{\epsilon}_{BDI}$, $\tilde{y}_{BCI} = \hat{y}_{BCI} + \hat{\epsilon}_{BCI}$ and $\tilde{y}_{BPI} = \hat{y}_{BPI} + \hat{\epsilon}_{BPI}$.
 Output: Final forecast results.

where $\hat{\mathbf{R}} \in \mathbb{R}^{n \times 1}$ is the corrected residual series. The final forecast is obtained by combining the DA-MTS prediction and the corrected residuals:

$$\hat{\mathbf{Y}} = \hat{\mathbf{Y}}_{DA-MTS} + \hat{\mathbf{R}}. \quad (10)$$

Prediction Performance Evaluation Criteria

This paper employs MAE, MAPE, and RMSE to assess the prediction performance of the models:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|, \quad (11)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|, \quad (12)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (13)$$

y_i and $\hat{y}_i$ represent the actual value and predicted value at time i , respectively, whereas n denotes the sample size of the test set.

Empirical Analysis**Data Description**

Dry Bulk Freight Rate Indices. This paper selects BDI, BCI, and BPI as the research objects. The dataset comprises monthly transactional data spanning from January 2000 to January 2024, a range determined by data availability. In 2003, the shipping market was affected by geopolitical situations, resulting in high volatility in freight rate indices, as shown in Figure 4a. In 2008, the financial crisis and global economic recession had a significant impact on the stability and price volatility of the freight market, as shown in Figure 4b. From 2009 to 2019, the freight rate fluctuations were relatively stable until the global spread of COVID-19, which led to supply chain disruptions and a decline in global freight demand,

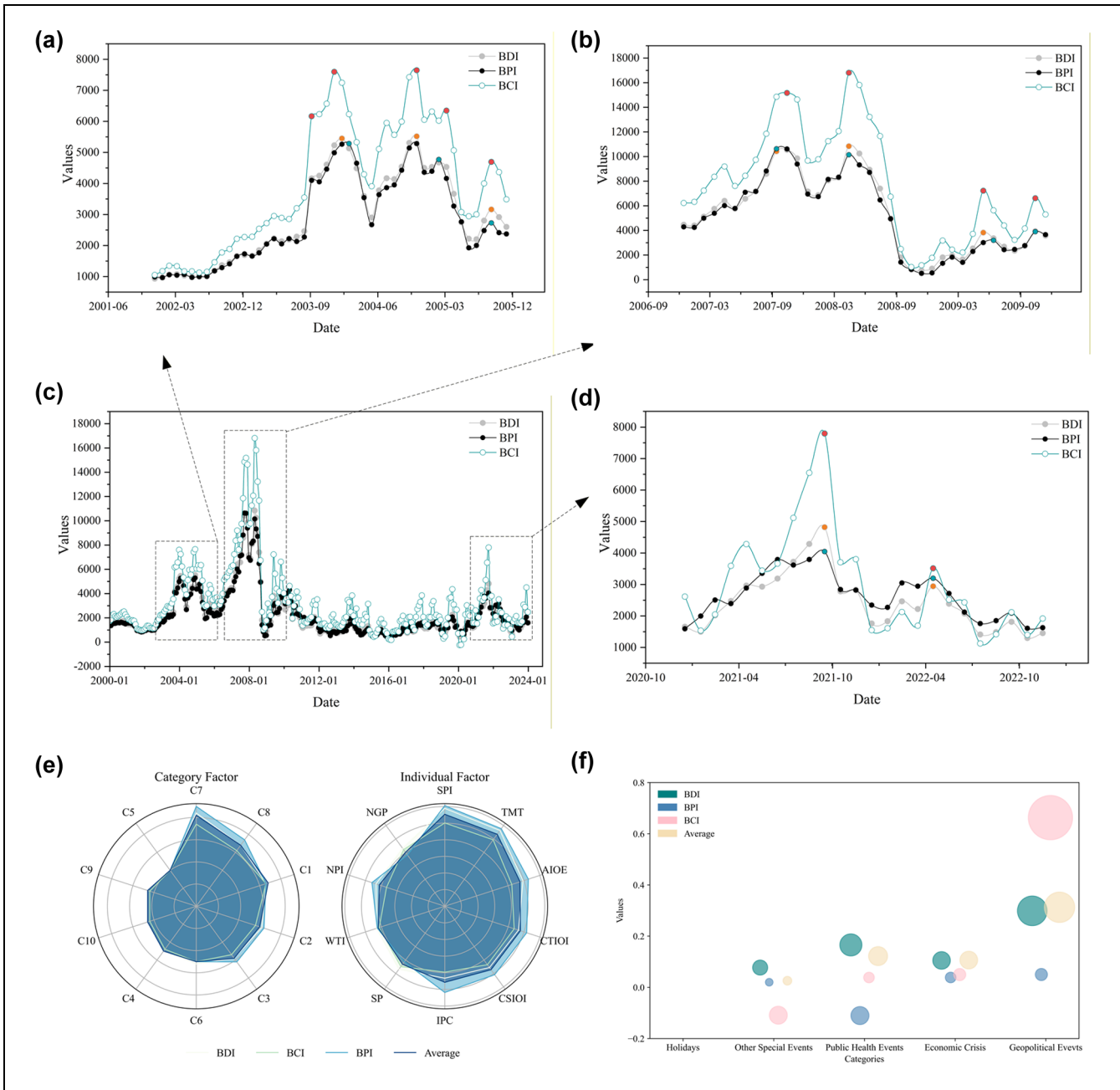


Figure 4. Freight rate fluctuations and correlation analysis: (a) Freight rate fluctuations under geopolitical tensions in 2003, (b) freight rate fluctuations under the impact of the 2008 financial crisis, (c) freight rate trends from January 2000 to January 2024, (d) freight rate fluctuations exhibit complex characteristics under the impact of varieties of factors, (e) correlation analysis between macro and micro exogenous variables and freight rates, and (f) correlation analysis between sudden events and freight rates.

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index. C* denotes category abbreviations, and C1-C11 represent different categories of indicators and unexpected events; see Table 8 for details. SPI = secondhand price index; TMT = TOPIX marine transportation; AIOE = Australia iron ore exports; CTIOE = China total iron ore imports; CSIOI = China seaborne iron ore imports; IPC = industrial production in China; SP = soybean price; WTI = West Texas Intermediate oil price; NPI = newbuilding price index; NGP = natural gas price.

causing freight rates to fluctuate sharply once again, as shown in Figure 4c. Figure 4d shows that BDI, BPI, and BCI exhibit complex characteristics such as non-stationarity, non-linearity, and high volatility. To

comprehensively evaluate the model's predictive performance, this paper divides the training sets and testing sets into ratios of 9:1, 8:2, 7:3, and 6:4. Descriptive statistics of the samples are shown in Table 7.

Table 7. Descriptive Statistics of the Samples

Category	Mean	Median	Maximum	Minimum	Standard deviation	Skewness	Kurtosis	Count
BDI	2240.7	1558.8	10843.6	306.9	1919.5	2.2	5.5	289
BCI	3180.1	2787.7	16808.1	243.1	2787.7	2.3	6.4	289
BPI	2237.8	1591.7	10631.2	324.2	1864.1	2.2	5.5	289

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index.

Table 8. Statistics of Macroeconomic Factor Categories and Quantities

Abbreviation	Category	Number	Abbreviation	Category	Number
C1	Commodity trade indicators	6	C7	Shipping trade indicators	6
C2	Global economic indicators	6	C8	Steel trade indicators	11
C3	Global industrial production indicators	32	C9	GPR indicators	112
C4	Energy price index	5	C10	U.S. stock market volatility indicators	45
C5	Public health indicators	1	C11	Unexpected events	5
C6	GEPU indicators	24			

Table 9. Statistics of Unexpected Event Categories

Category	Abbreviation	Sample	Total time
Public health events	PHE	SARS, COVID-19 et al.	90 months
Geopolitical events	GE	Iraq War, Russia-Ukraine conflict et al.	128 months
Holidays	HOL	Chinese New Year, U.S. Christmas et al.	96 months
Economic crisis	EC	2008 financial crisis, 2015 Chinese stock market volatility et al.	48 months
Other special events	OSE	IMO leadership election, Black Sea Agreement signing et al.	20 months

Multidimensional Factors Database. To enhance prediction accuracy and comprehensively consider the impact of various factors on freight rates, this paper constructs a multidimensional factor database, which includes 4,281 time series data from twelve public datasets, 248 exogenous variables from five non-public commercial datasets, and five self-building binary sequence indicators for sudden events. We extract the time points of unexpected events (such as geopolitical events, economic crises, holidays, and public health events) from January 2000 to January 2024. The value is set to 1 at the time points when these events occur and 0 otherwise. These unexpected event time points are encoded as binary sequence data, and the MIC algorithm is used to quantify the correlation between freight rates and unexpected events. The types and numbers of market factor indicators are exhibited in Table 8, the categories and time statistics of unexpected events are displayed in Table 9, and partly micro factors and their corresponding abbreviations are shown in Table 10.

Data Processing. To eliminate the impact of different data scales between various features, this paper uses the Min-

Max normalization method to process various indicators, as shown in Equation (14). Min-Max normalization scales all variables to the same numerical range (0 to 1), reducing the influence of scale differences and facilitating feature extraction during model training. This approach is particularly beneficial for heterogeneous datasets, as it ensures consistency across domains (shipping, finance, and commodities) and avoids potential overfitting caused by extreme values in specific domains. The domain differences in economic time series (e.g., volatility, periodicity, and trends) may introduce distributional bias during unified normalization. The self-attention mechanism in the Transformer architecture automatically captures key dependencies and domain-specific features during training. The self-attention mechanism enables the model to focus on the most relevant features across different domains, mitigating potential biases introduced by preprocessing. Furthermore, we conducted comparative experiments to evaluate the impact of different normalization methods (Min-Max, Z-score, Quantile, and Robust Scaling) on model performance. It is important to note that for each normalization method, the corresponding transformation was consistently applied during

Table 10. Abbreviations and Categories of Specific Micro Factors

Indicator	Abbreviation	Category
Cotton price index	CPI	C1
China's soybean imports	CSI	
Inflation indicator USA	IIU	C2
Inflation indicator OECD	II OECD	
Inflation indicator South Korea	IIK	C3
USA dollar index	UDI	
Industrial production China	IPC	
Industrial production Latin America	IPLA	
Industrial production India	IPI	
Industrial production Brazil	IPB	
Industrial production Poland	IPP	
Industrial production Southeast Asia average	IPSEAA	
Industrial production Mexico	IPM	
Industrial production Russia	IPR	
Industrial production New Zealand	IPN	C5
Infectious disease equity market volatility tracker	ID EMVT	
TOPIX marine transportation	TMT	C7
Secondhand price index	SPI	
Brazil iron ore exports	BIOE	C8
Ukraine iron ore exports	UIOE	
Geopolitical risk historical index of Poland	GPRHC POL	C9
Labor regulations equity market volatility tracker	LR EMVT	
Macro real estate markets equity market volatility tracker	MREM EMVT	C10
Petroleum markets equity market volatility tracker	PM EMVT	
Labor disputes equity market volatility tracker	LD EMVT	C11
Geopolitical events	GE	

Table 11. Comparison of Different Normalization Methods on BDI, BCI, and BPI datasets using MAE, RMSE, and MAPE as Evaluation Metrics

Normalization method	BDI			BCI			BPI		
	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %
Min.–Max. normalization	0.19	0.23	6.24	0.26	0.30	8.08	0.18	0.22	5.84
Z-score normalization	0.22	2.25	6.74	0.28	0.37	8.51	0.24	0.31	6.17
Quantile normalization	0.21	0.28	6.50	0.27	0.35	8.35	0.20	0.26	6.00
Robust scaling	0.23	0.29	6.80	0.29	0.38	8.60	0.25	0.32	6.25

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; MAE = mean absolute error; RMSE = root mean square error; MAPE = mean absolute percentage error.

both the pretraining and fine-tuning stages to ensure alignment across the entire training pipeline. The results, detailed in Table 11, show that Min-Max normalization achieves the best balance between computational efficiency and prediction accuracy.

$$\hat{X} = \frac{X - X_{min}}{X_{max} - X_{min}}, \quad (14)$$

X_{min} and X_{max} are the minimum and maximum value in the series, respectively; $\hat{X}$ is the normalized result.

Analysis of the Correlation between Exogenous Variables and Freight Rates

We operate the MIC algorithm to explore the correlation between exogenous variables and freight rates. At the macro level, shipping trade indicators correlate more with freight rates than other categories, whereas public health indicators have the lowest correlation with freight rates. The secondhand price index correlates with freight rates at the micro level, followed by the TOPIX marine transportation. Among them, indicators highly correlated with freight

Table 12. Correlation between Freight Rates and Macroeconomic Factors

Category of macroeconomic factors	Abbreviation	BDI	BCI	BPI	Average
Shipping trade indicators	C7	0.4200	0.3705	0.4451	0.4128
Steel trade indicators	C8	0.3400	0.3124	0.3686	0.3419
Commodity trade indicators	C1	0.3300	0.3320	0.3341	0.3351
Global economic indicators	C2	0.3100	0.2806	0.3208	0.3036
Global industrial production indicators	C3	0.2900	0.2664	0.3063	0.2887
GEPU indicators	C6	0.2600	0.2494	0.2495	0.2545
Energy price index	C4	0.2500	0.2442	0.2492	0.2491
U.S. stock market volatility indicators	C10	0.2300	0.2192	0.2320	0.2284
GPR indicators	C9	0.2305	0.2200	0.2303	0.2269
Public health indicators	C5	0.2020	0.2002	0.2000	0.2007

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index.

Table 13. Correlation between Freight Rates and Microeconomic Factors

Category	Abbreviation	BDI	BCI	BPI	Average
Secondhand price index	SPI	0.5886	0.5010	0.6052	0.5531
TOPIX marine transportation	TMT	0.5637	0.4938	0.5766	0.5352
Australia iron ore exports	AIOE	0.4935	0.4230	0.5273	0.4752
China total iron ore imports	CTIOE	0.4652	0.4383	0.5166	0.4775
China seaborne iron ore imports	CSIOI	0.4561	0.4350	0.5091	0.4721
Industrial production China	IPC	0.4331	0.3964	0.5185	0.4575
Soybeans price	SP	0.4678	0.4492	0.4139	0.4315
WTI oil price	WTI	0.4308	0.4138	0.4260	0.4199
Newbuilding price index	NPI	0.4341	0.3677	0.4598	0.4138
Natural gas price	NGP	0.4299	0.4209	0.3808	0.4008

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index.

rates exhibit similar long-term dependencies and stable variation trends as freight rates, whereas macro factors with lower correlations, such as public health indicators and GPR indicators, show more sudden changes. Overall, the correlations of all variables are significant. The comparison of the correlations between the freight rate indices and market factors are shown in Figure 4e, the correlations between macro factors and freight rates are shown in Table 12, and the top ten micro factors exhibiting the highest correlations with freight rates are shown in Table 13.

Analysis of the Correlation between Sudden Events and Freight Rates

This paper selects holidays as the baseline events, progressively adds other special events, public health events, economic crises, and geopolitical events, and calculates the MIC values between the new series formed by each type of event and the freight rates. The results show that MIC values exhibit an upward trend, indicating that various events have a cumulative effect on freight rates. Specifically, when counting the time series of geopolitical events, the MIC value's growth rate is the most

significant, reaching 31.27%, followed by public health events and economic crises, with 12.21% and 10.65%, respectively. This indicates that geopolitical events and economic crises affect freight rate fluctuations more than holidays and other special events. Based on these conclusions, it is recommended that participants in the shipping market closely monitor the trends in GPR, public health, and global economic indicators. The comparison of the correlations between sudden events and freight rates is shown in Figure 4f, and the MIC values are shown in Table 14.

Freight Rate Prediction

DA-MTS Prediction. This paper conducts a comparative analysis of the prediction results of Transformer (baseline), MTS (pre-trained), and DA-MTS (fine-tuned). The results show that Transformer has the most significant prediction error and the worst overall performance, whereas DA-MTS can significantly reduce model error and has the best overall performance. As detailed in Table 15, the model achieves optimal predictive performance when the num_layers is set to twelve. The DA-

Table 14. DA-MTS Forecast Freight Rates Error Evaluation Metrics

Category	BDI-MIC	BPI-MIC	BCI-MIC	AVERAGE	Growth rate, %
HOL	0.1986	0.2022	0.1909	0.1972	
HOL-OSE	0.2139	0.1802	0.2129	0.2023	2.59
HOL-OSE-PHE	0.2494	0.1871	0.2446	0.2270	12.21
HOL-OSE-PHE-EC	0.2756	0.1964	0.2816	0.2512	10.65
HOL-OSE-PHE-EC-GE	0.3580	0.3267	0.3046	0.3297	31.27

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; DA-MTS = domain-adapted maritime time series prediction model; MIC = maximum mutual information coefficient; HOL = holidays; OSE = other special events; PHE = public health events; EC = economic crisis; GE = geopolitical events. HOL-OSE, HOL-OSE-PHE, HOL-OSE-PHE-EC, and HOL-OSE-PHE-EC-GE denote composite categories formed by combining the corresponding event types.

Table 15. Performance Comparison of DA-MTS with Different Numbers of Layers (num_layers) across BDI, BCI, and BPI Metrics

Models	BDI			BCI			BPI		
	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %
DA-MTS (num_layers = 8)	0.23	0.28	7.50	0.3	0.35	9.50	0.22	0.27	7.20
DA-MTS (num_layers = 10)	0.21	0.26	6.80	0.28	0.33	8.70	0.20	0.25	6.40
This paper (num_layers = 12)	0.19	0.23	6.24	0.26	0.3	8.08	0.18	0.22	5.84
DA-MTS (num_layers = 14)	0.20	0.25	6.50	0.27	0.32	8.30	0.19	0.24	6.10
DA-MTS (num_layers = 16)	0.22	0.27	6.90	0.29	0.34	8.90	0.21	0.26	6.50

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; MAE = mean absolute error; MAPE = mean absolute percentage error; DA-MTS = domain-adapted maritime time series prediction model; RMSE = root mean square error.

MTS prediction fitting effect for BDI is shown in Figure 5, for BCI in Figure 6, and for BPI in Figure 7. The evaluation metrics for model prediction errors are shown in Figure 8. The error evaluation metrics are provided in Tables 16–18.

LSTM Prediction. First, the GWO algorithm is used to optimize the number of hidden units, maximum training epochs, activation function, and dropout rate of the LSTM model. Based on the hyperparameters above, the LSTM model is subsequently constructed.

The results show that the LSTM model, after parameter optimization, can accurately identify the short-term fluctuation characteristics of freight rates that DA-MTS failed to capture when fitting the residuals of the DA-MTS freight rate forecast results. The residual forecast fitting of the LSTM model for BDI is shown in Figure 9, for BCI in Figure 10, and for BPI in Figure 11.

DA-MTS+LSTM Prediction. The results show that the errors of the hybrid model's forecast results are further reduced based on DA-MTS. When forecasting BDI, the metrics show significant improvements: MAE decreases by 15.79%, RMSE by 8.70%, and MAPE by 15.71%. Similarly, for BCI forecast, MAE is decreased by 19.23%, RMSE by 10%, and MAPE by 17.45%. As for BPI, the

reductions are 11.11% in MAE, 9.09% in RMSE, and 12.33% in MAPE. This indicates that the model effectively captures short-term fluctuations in freight rates. The fitting results of the hybrid model for forecasting BDI are shown in Figure 12, for forecasting BCI in Figure 13, and for forecasting BPI in Figure 14. The comparison of the error evaluation metrics between the hybrid model and DA-MTS for forecasting BDI, BCI, and BPI is shown in Figure 15. The error evaluation metrics are provided in Tables 19–21.

Analysis of Factor Importance

Long-Term. From a long-term perspective, shipping trade indicators are the most significant for forecasting freight rates, followed by commodity trade, global economic, and public health indicators. In contrast, GEPU, GPR, and unexpected events have a minor long-term impact on freight rate predictions. These conclusions indicate that the balance of supply and demand in shipping, public health conditions, the supply and demand in the commodity market, and economic growth trends are key factors influencing freight rates. At the same time, policy changes and unexpected events have a more negligible impact on the long-term trend of freight rates. The importance of macro factors is shown in Figure 16a.

The importance of micro factors is shown in Figure 16c. The TOPIX marine transportation,



Figure 5. The performance comparison of DA-MTS in BDI prediction under four different partitioning forms.

Note: BDI = baltic dry index; DA-MTS = domain-adapted maritime time series prediction models; MTS = maritime time series prediction model.

secondhand price index, industrial production China, and US dollar index are significantly more critical for forecasting freight rate indices than other micro factors. This indicates that the performance of the shipping industry's stocks, dynamics of the secondhand ship market, China's industrial production conditions, and the international influence of dollars are critical drivers of freight rate fluctuations.

Short-Term. The importance of macro factors is shown in Figure 16b. The energy price index, global industrial production indicators, and GPR indicators significantly influence short-term freight rates. Excluding the energy price index raises MAE for BDI, BCI, and BPI by 11.39%, 6.11%, and 7.75%, respectively. Similarly, omitting GPR increases MAE for these indices by 12.37%, 4.89%, and 6.39%. Excluding global industrial production indicators raises MAE for BDI, BCI, and

BPI by 10.97%, 4.47%, and 6.89%, respectively. In contrast, shipping trade and global economic indicators have a smaller impact. For example, removing shipping trade indicators reduces MAE for BDI and BPI by 0.83% and 0.81%. Meanwhile, US stock market volatility indicators, GEPU, and unexpected events strongly affect BPI. Excluding GEPU raises MAE for BPI by 7.003%, and for BDI and BCI by 6.87% and 3.43%.

Figure 16d reveals the micro factors that impact freight rate indices. Geopolitical events, labor regulations equity market volatility, Ukraine iron ore exports, industrial production Russia, public health events, China soybean imports, and inflation indicator USA significantly influence freight rates. For instance, excluding geopolitical events raises the MAE of BDI, BCI, and BPI by 2.76%, 2.08%, and 3.32%, respectively, showing their direct impact on global shipping. Excluding industrial production Poland increases the MAE of BDI, BCI, and BPI by 2.78%, 1.63%, and 3.2%, respectively,

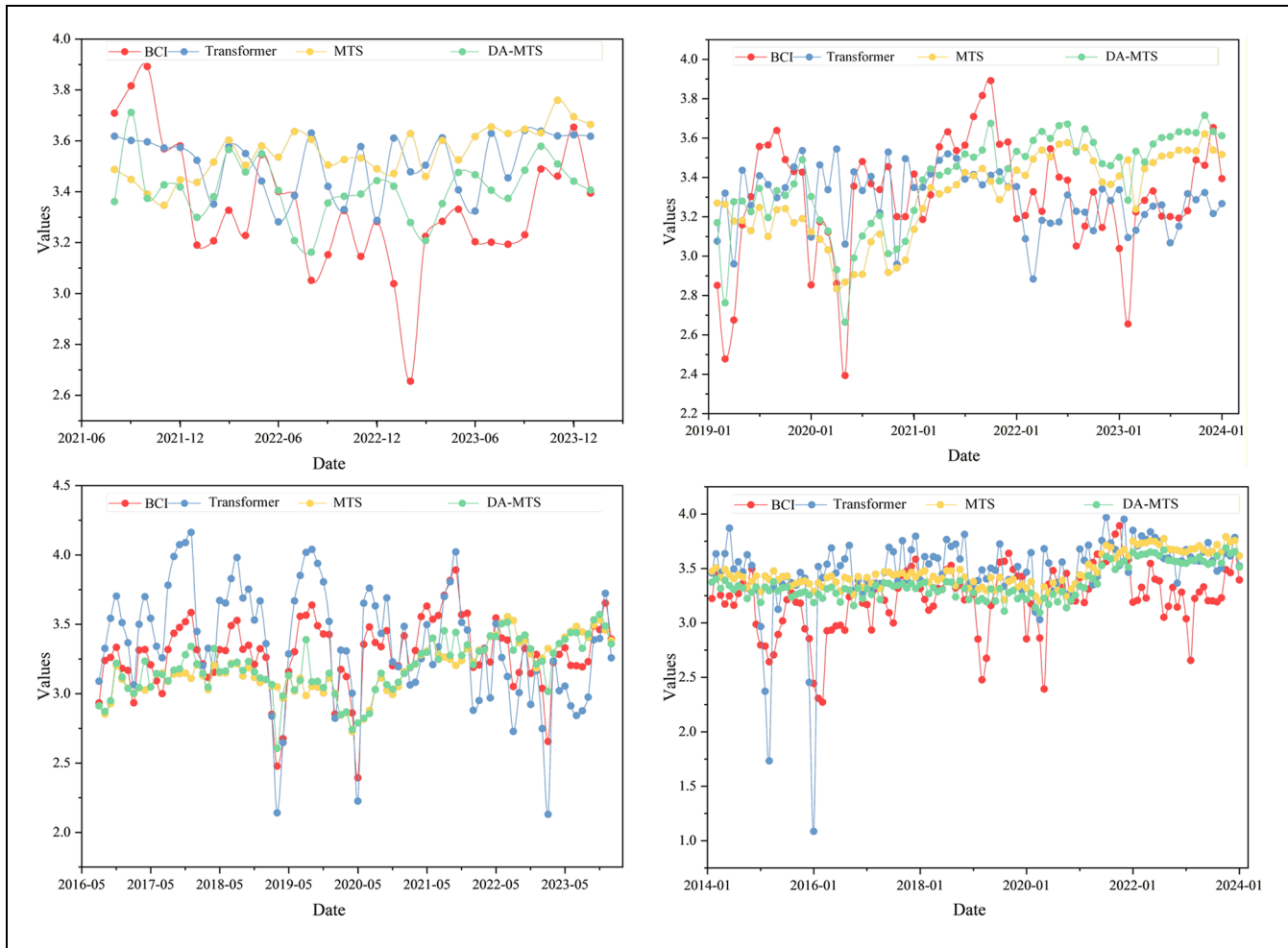


Figure 6. The performance comparison of DA-MTS in BCI prediction.

Note: BCI = baltic capesize index; DA-MTS = domain-adapted maritime time series prediction model; MTS = maritime time series prediction model.

highlighting its importance for European and global shipping demand. Sensitivity differences of BDI, BCI, and BPI to micro factors are evident in three areas: (1) The labor regulations equity market volatility tracker significantly affects BDI. (2) China soybean imports and industrial production Russia affect BCI. (3) BPI is sensitive to industrial production Poland, Ukraine iron ore exports, geopolitical events, public health events, and inflation indicator USA. The comparison tables of the importance of long-term and short-term macro and micro factors for BDI, BCI, and BPI can be found in Tables 22–27.

Analysis of the Importance of Unexpected Events. Figure 16, *e* and *f*, shows the importance of five unexpected events on freight rate fluctuations from long-term and short-term perspectives. The main conclusions are as follows: (1) Geopolitical events and economic crises have a more

significant impact on freight rates than other sudden events. (2) Unexpected events such as IMO leadership elections, the signing of the Black Sea Agreement, policy announcements, and national leader elections have a significantly lower impact on freight rates than geopolitical events, holidays, and public health events. (3) Holidays have a more significant short-term impact on freight rates than public health events. (4) Geopolitical events like the Iraq War and the Russia–Ukraine conflict, public health events like SARS and COVID-19, and economic crises like the 2008 financial crisis and 2015 Chinese stock market fluctuations have a significantly higher impact on BPI than BCI. However, holidays, such as Chinese New Year and Christmas, and sudden events like IMO leadership elections and the signing of Black Sea Agreement have a more significant short-term impact on BCI. The comparison table of importance of long-term and short-term for unexpected events with other existing models can be found in Table 28.

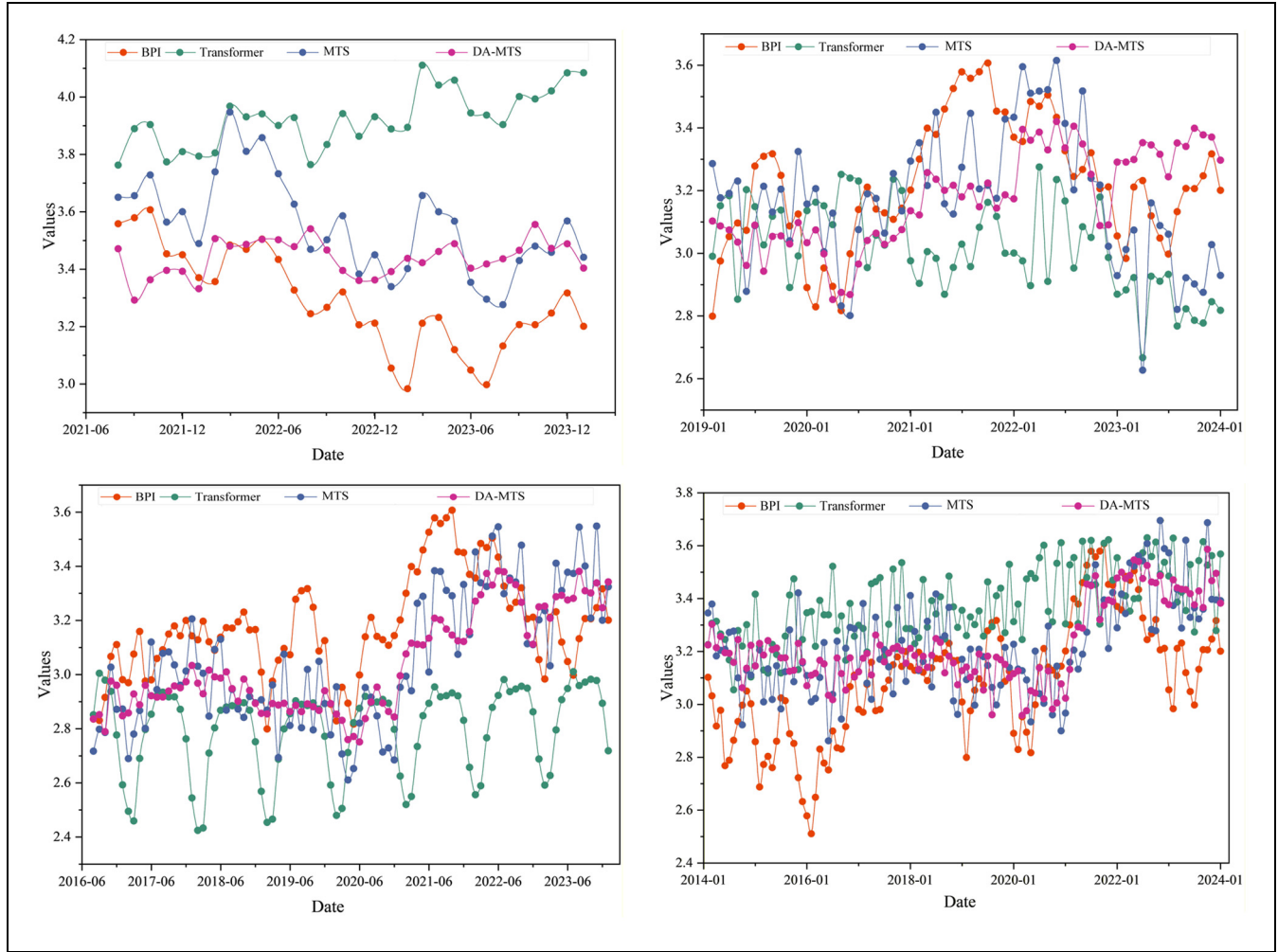


Figure 7. The performance comparison of DA-MTS in BPI prediction.

Note: BPI = baltic panamax index; DA-MTS = domain-adapted maritime time series prediction model; MTS = maritime time series prediction model.

In this section we conduct a comparative analysis of the prediction performance of the DA-MTS+LSTM model against SMs represented by ARIMA, ML models represented by ANN, and forecast models based on Transformer architecture represented by Autoformer. The objective is to verify the significant advantages of the DA-MTS+LSTM model in the task of dry bulk freight rate prediction. With regard to forecast accuracy, the DA-MTS+LSTM model significantly outperforms traditional SMs and the latest DL methods, especially in forecasting BDI and BPI. The performance comparison between DA-MTS+LSTM model and SMs, ML models, and time series forecast models based on Transformer architecture is shown in Table 29. The visual comparisons of model performance are

shown in Figure 17a. A positive value indicates that the model's error is higher than the proposed model, whereas a negative value indicates that the error is lower. The absolute value represents the percentage difference in error compared with the model proposed in this paper. Furthermore, the results of the ablation study are provided in Table 30.

Figure 17b shows the visualization results of the average MAE, RMSE, and MAPE for BDI, BCI, and BPI forecasts by SMs, ML, and time series prediction models based on Transformer architecture. Overall, time series forecast models based on Transformer architecture outperform SMs and ML methods with regard to comprehensive forecast performance, and ML methods generally perform better than SMs.

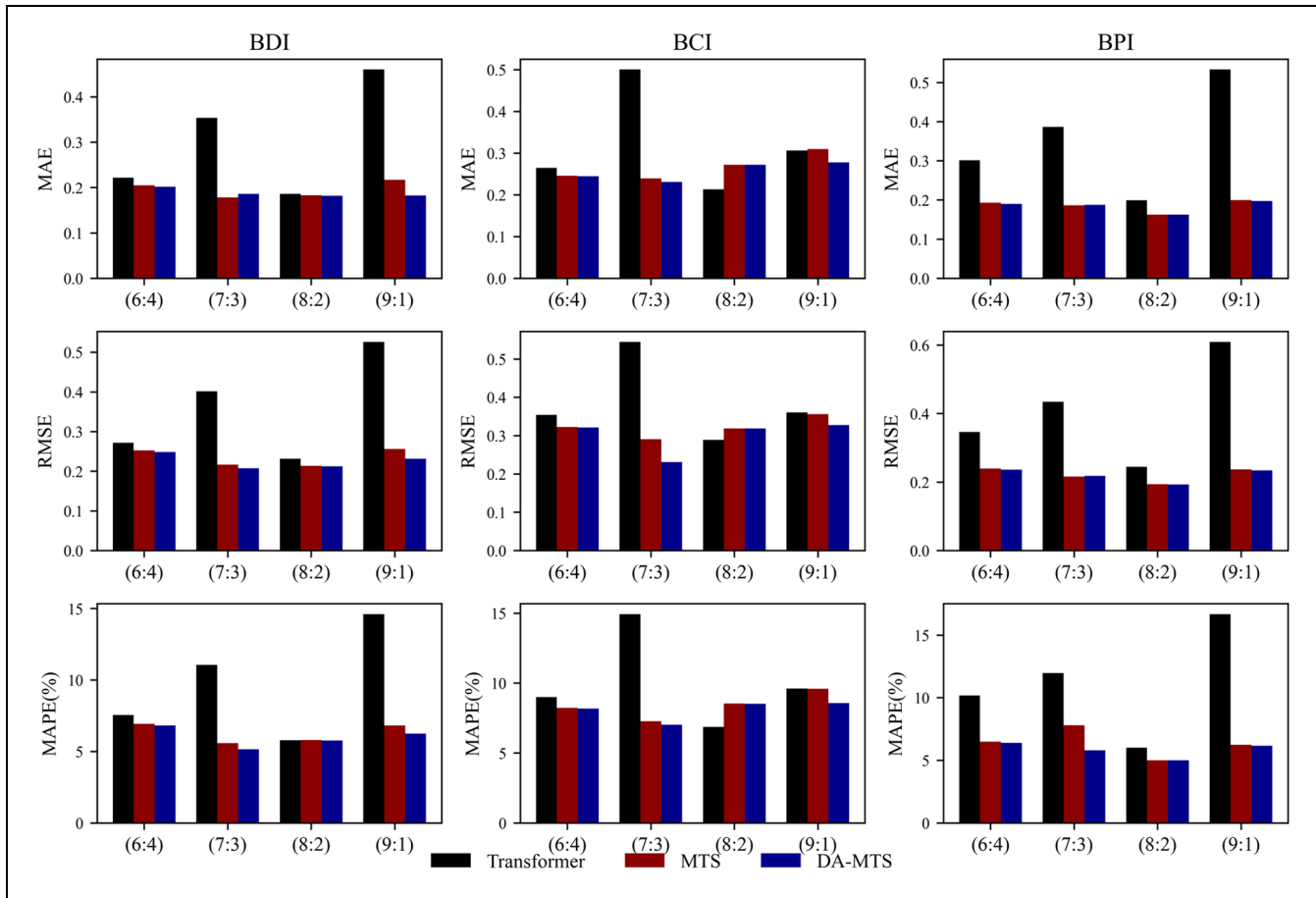


Figure 8. The error metrics of DA-MTS in dry bulk freight rate prediction.

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; DA-MTS = domain-adapted maritime time series prediction model.

Table 16. DA-MTS Forecast BDI Error Evaluation Metrics

BDI	Category	MAE	RMSE	MAPE, %
(6:4)	Transformer	0.2217	0.2718	7.5452
	MTS	0.2050	0.2526	6.9247
	DA-MTS	0.2017	0.2482	6.8076
(7:3)	Transformer	0.3540	0.4015	11.0546
	MTS	0.1782	0.2166	5.5773
	DA-MTS	0.1715	0.2075	5.1532
(8:2)	Transformer	0.1859	0.2317	5.7892
	MTS	0.1828	0.2136	5.8034
	DA-MTS	0.1816	0.2125	5.7621
(9:1)	Transformer	0.4603	0.5261	14.6028
	MTS	0.2171	0.2562	6.8200
	DA-MTS	0.1824	0.2316	6.2400

Note: BDI = baltic dry index; DA-MTS = domain-adapted maritime time series prediction model; MTS = maritime time series prediction model; MAE = mean absolute error; RMSE, root mean square error; MAPE, mean absolute percentage error.

Table 17. DA-MTS Forecast BCI Error Evaluation Metrics

BCI	Category	MAE	RMSE	MAPE, %
(6:4)	Transformer	0.2646	0.3539	9.0045
	MTS	0.2458	0.3222	8.2223
	DA-MTS	0.2448	0.3209	8.1857
(7:3)	Transformer	0.5004	0.5446	14.9301
	MTS	0.2394	0.2905	7.2801
	DA-MTS	0.2308	0.2308	7.0256
(8:2)	Transformer	0.2134	0.2885	6.8726
	MTS	0.2722	0.3187	8.5301
	DA-MTS	0.2719	0.3186	8.5153
(9:1)	Transformer	0.3064	0.3602	9.6079
	MTS	0.3099	0.3563	9.6000
	DA-MTS	0.2775	0.3273	8.5800

Note: BCI = baltic capesize index; MTS = maritime time series prediction model; MAE = mean absolute error; RMSE = root mean square error; MAPE = mean absolute percentage error; DA-MTS = domain-adapted maritime time series prediction model.

Table 18. DA-MTS Forecast BPI Error Evaluation Metrics

BPI	Category	MAE	RMSE	MAPE, %
(6:4)	Transformer	0.3014	0.3459	10.1814
	MTS	0.1927	0.2397	6.4914
	DA-MTS	0.1897	0.2361	6.3882
(7:3)	Transformer	0.3866	0.4344	11.9766
	MTS	0.1862	0.2157	0.7799
	DA-MTS	0.1872	0.2178	5.8070
(8:2)	Transformer	0.1989	0.2446	6.0031
	MTS	0.1626	0.1937	5.0042
	DA-MTS	0.1623	0.1932	5.0013
(9:1)	Transformer	0.5332	0.6091	16.6660
	MTS	0.1995	0.2367	6.2300
	DA-MTS	0.1975	0.2342	6.1600

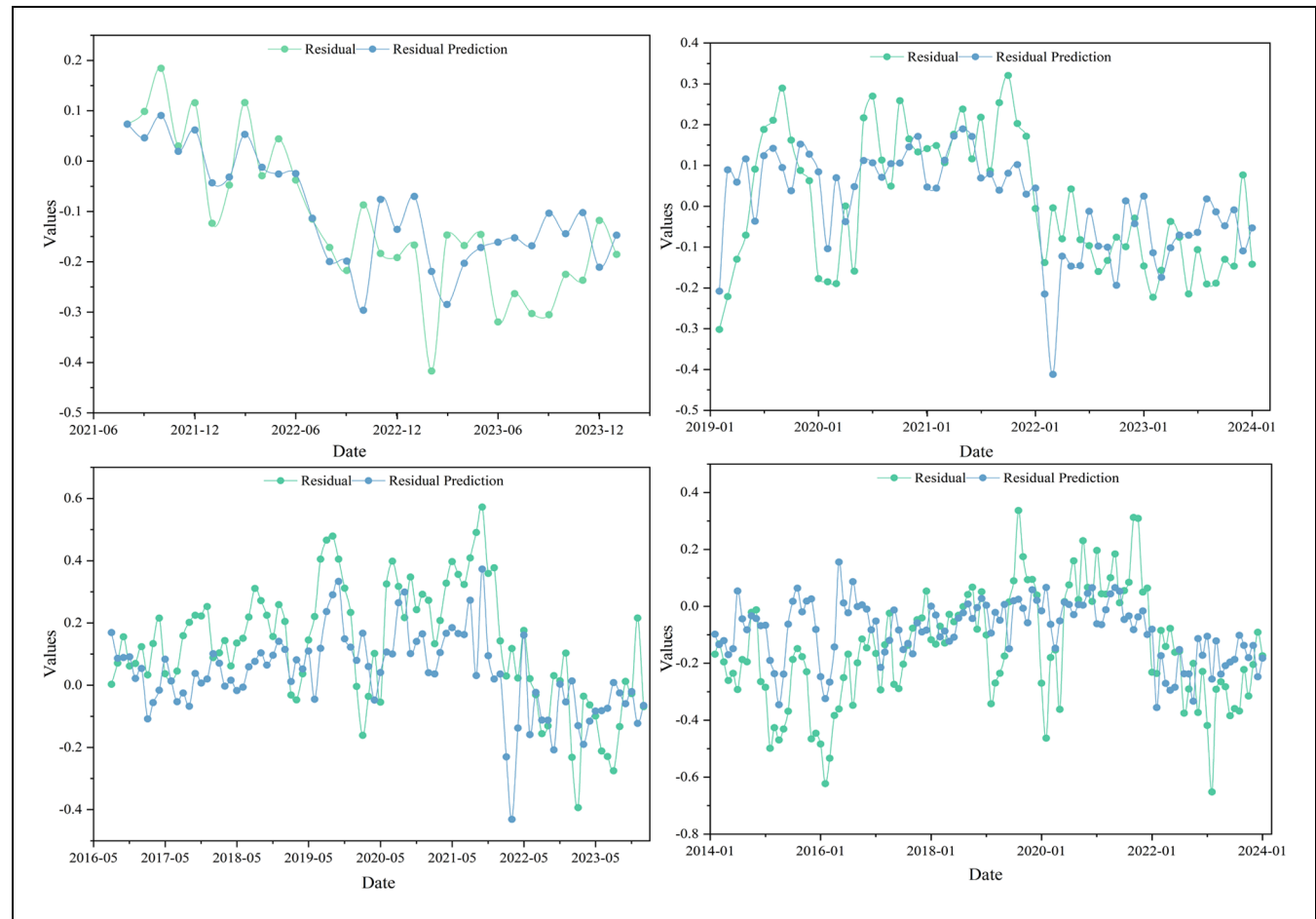
Note: BPI = baltic panamax index; MTS = maritime time series prediction model; MAE = mean absolute error; RMSE = root mean square error; MAPE = mean absolute percentage error; DA-MTS = domain-adapted maritime time series prediction model.

Conclusion and Prospects

Conclusion

Based on the DA-MTS+LSTM model, this paper establishes a hybrid freight rate forecast model that considers exogenous variables and unexpected events and draws the following conclusions:

- (1) \The proposed MTS framework represents the first pre-trained model for time series prediction across integrated domains such as shipping, finance, and commodities. Building on this foundation, DA-MTS is the first fine-tuned model specifically tailored for forecasting dry bulk shipping price indices. Additionally, the framework provides a comprehensive workflow (Figure 18), spanning from data collection and feature selection to model training, validation, and industry

**Figure 9.** The residual prediction results for BDI under four different partitioning forms.

Note: BDI = baltic dry index.

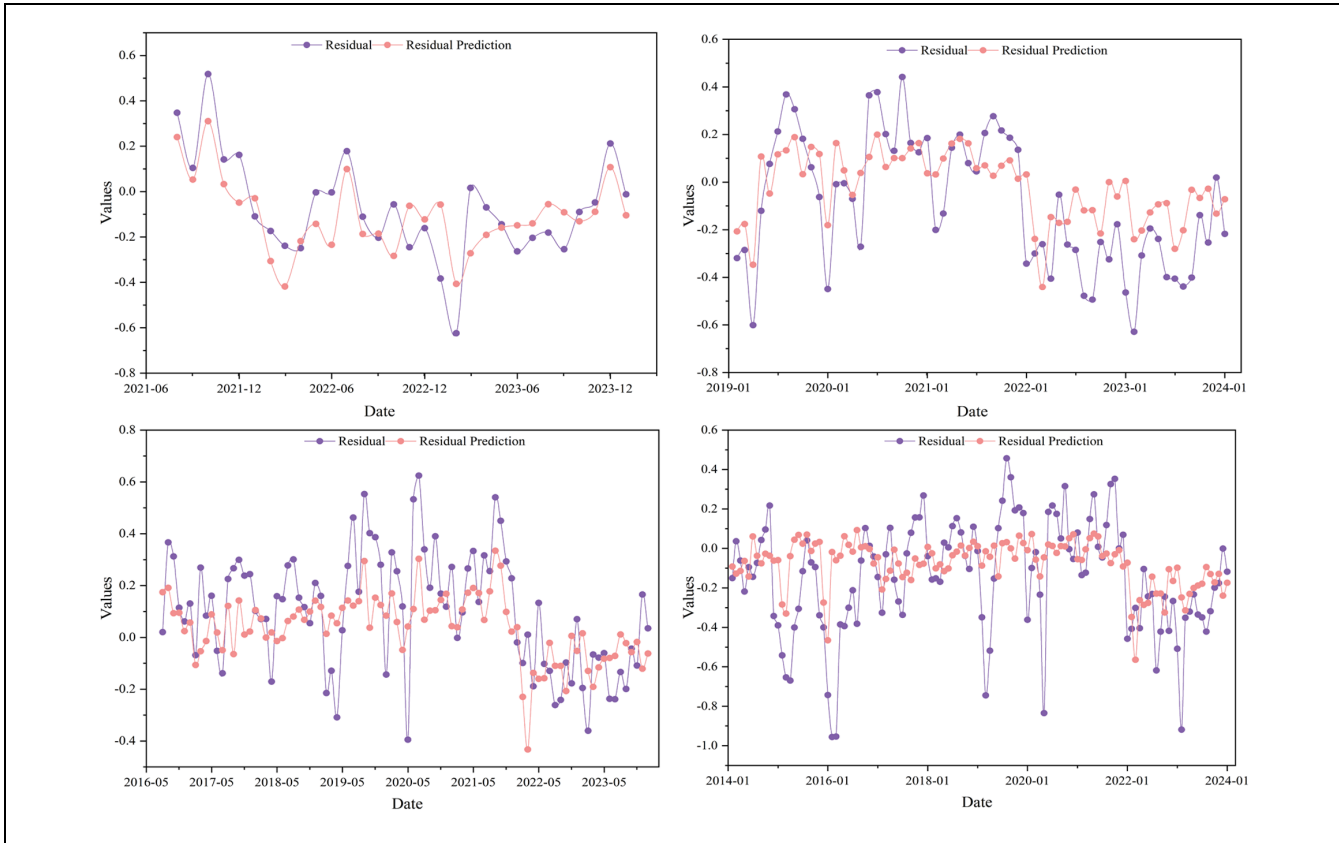


Figure 10. The residual prediction results for BCI under four different partitioning forms.

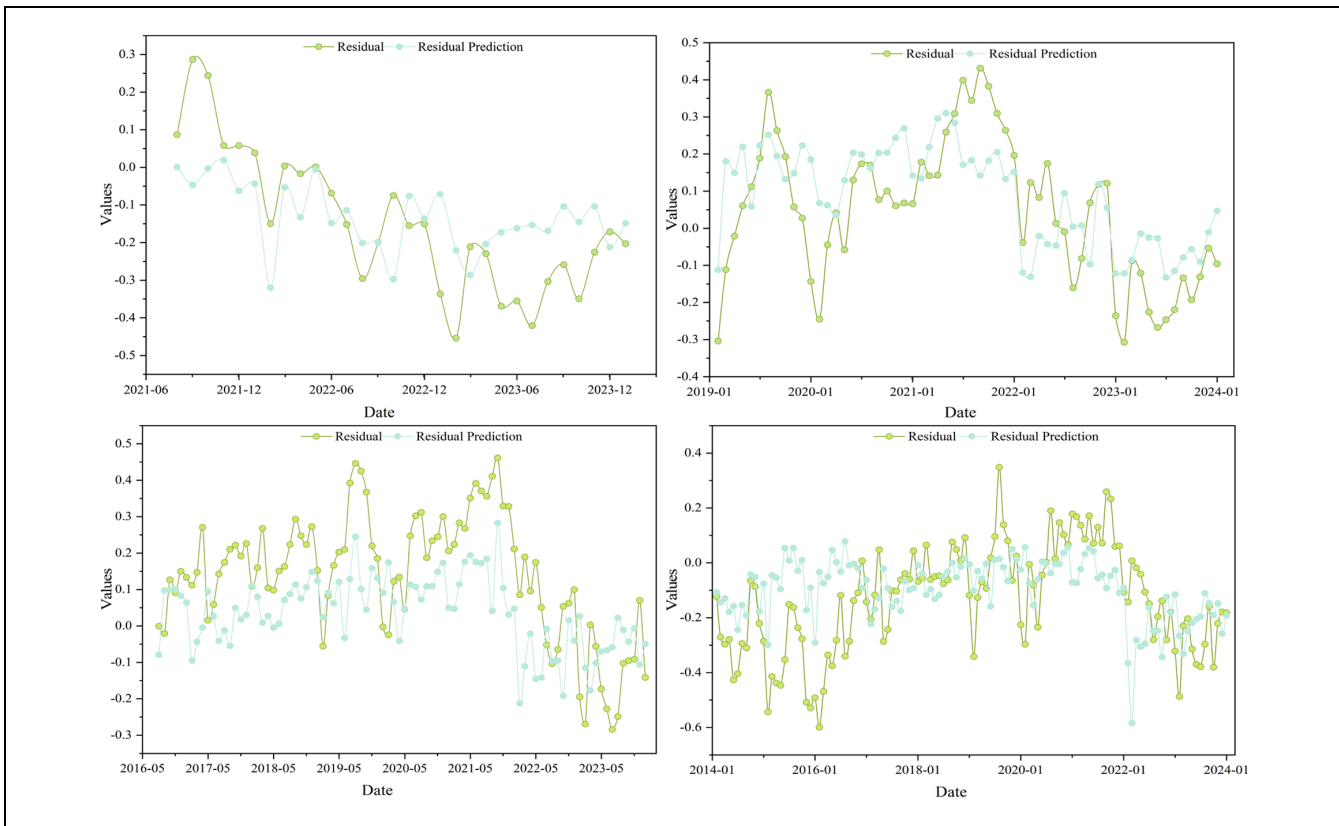


Figure 11. The residual prediction results for BPI under four different partitioning forms.

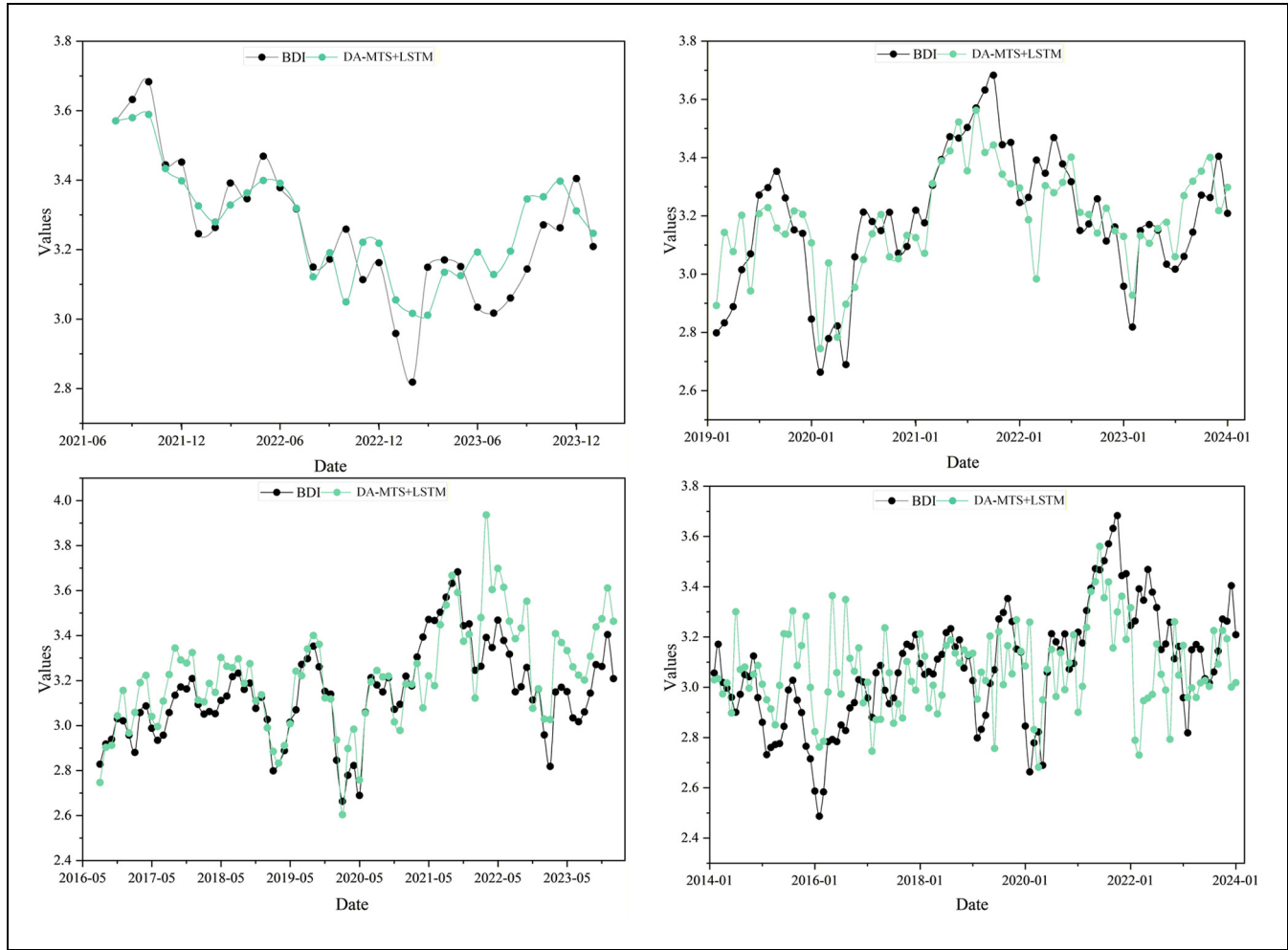


Figure 12. The prediction results of the hybrid model proposed in this paper (DA-MTS+LSTM) for BDI under four different partitioning forms, respectively.

BDI = baltic dry index; DA-MTS = domain-adapted maritime time series prediction model; LSTM = long short-term memory.

- integration, offering a systematic solution to address complex prediction challenges.
- (2) By combining DA-MTS's ability to capture long-term freight rate fluctuations with LSTM's strength in identifying short-term freight rate changes, the DA-MTS+LSTM model can accurately grasp dynamic characteristics of freight rates, significantly improving forecast accuracy and providing strong support for the operational decision-making of shipping companies.
- (3) Through comparative analysis of the forecast results of SMs such as ARIMA, ML models like ANN, and DL methods based on the Transformer architecture like Autoformer, it is confirmed that the DA-MTS+LSTM model adopted in this paper has the highest forecast accuracy in freight rate forecasting.
- (4) With regard to analyzing factors affecting freight rates, previous studies have mainly focused on GPR, public health events, EPU, and crude oil price. This paper comprehensively compares, analyzes, and quantifies the impact of exogenous variables and unexpected events on freight rates from both long-term and short-term perspectives. The conclusions of this study are generally consistent with previous research. The findings of this study provide essential reference points for decision-makers in related fields and offer a new perspective for understanding the mechanisms of freight rate fluctuations and new methods for solving forecast tasks in the maritime domain. Comparisons of the study's conclusions with existing research on factors affecting freight

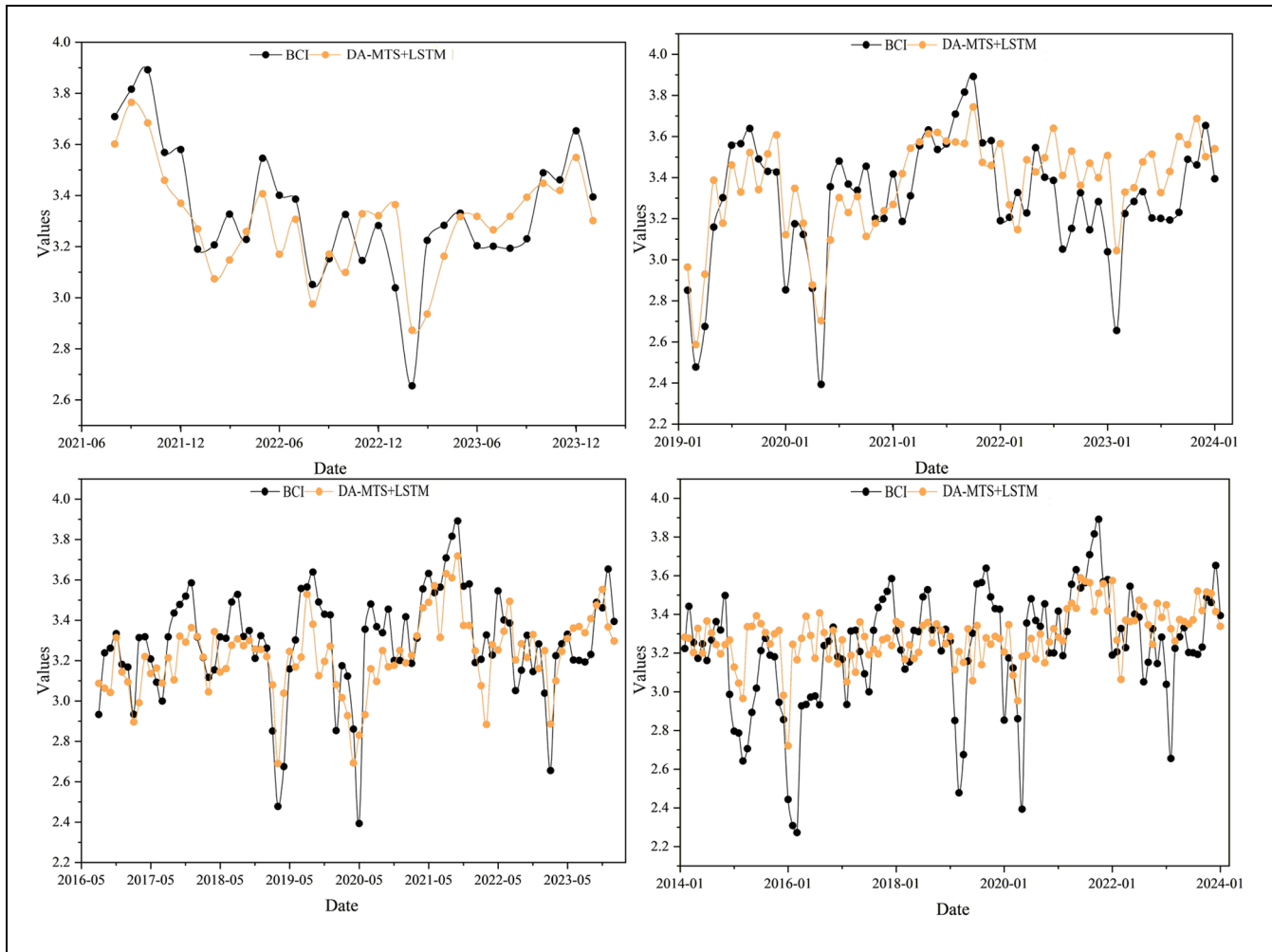


Figure 13. The prediction results of the hybrid model proposed in this paper (DA-MTS+LSTM) for Baltic Capesize Index (BCI) under four different partitioning forms, respectively.

rates and new findings of this study are shown in Tables 31 and 32, respectively.

Prospects

Based on the current research findings, further exploration can be conducted in the following directions.

- (1) MTS has transfer learning capabilities and can be extended to various fields, such as financial and energy markets.
- (2) Develop an intelligent interactive risk management system. The DA-MTS forecasting results are input into an NLP model to generate detailed solutions or recommendations for different user groups. The generated recommendations can be integrated into an intelligent interactive system, allowing different users to directly obtain them through an intelligent interface or voice

interaction for risk management and scientific decision-making.

- (3) Cross-domain data integration and text data feature extraction: With the advancement of big data technology and AI quantification, incorporating text data—such as social media sentiment analysis and news event impact assessments—into models as exogenous variables enhances the accuracy of freight rate forecasting. Additionally, it enables the analysis of the specific effects of news and social media on freight rate fluctuations.

Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: Yuping Luo, Rui Zhang, Zhiyuan Liu, Wenyang Wang, Yajuan Xiang, Jibin Zhou; data collection: Yuping Luo, Rui Zhang, Wenyang Wang, Yajuan Xiang,

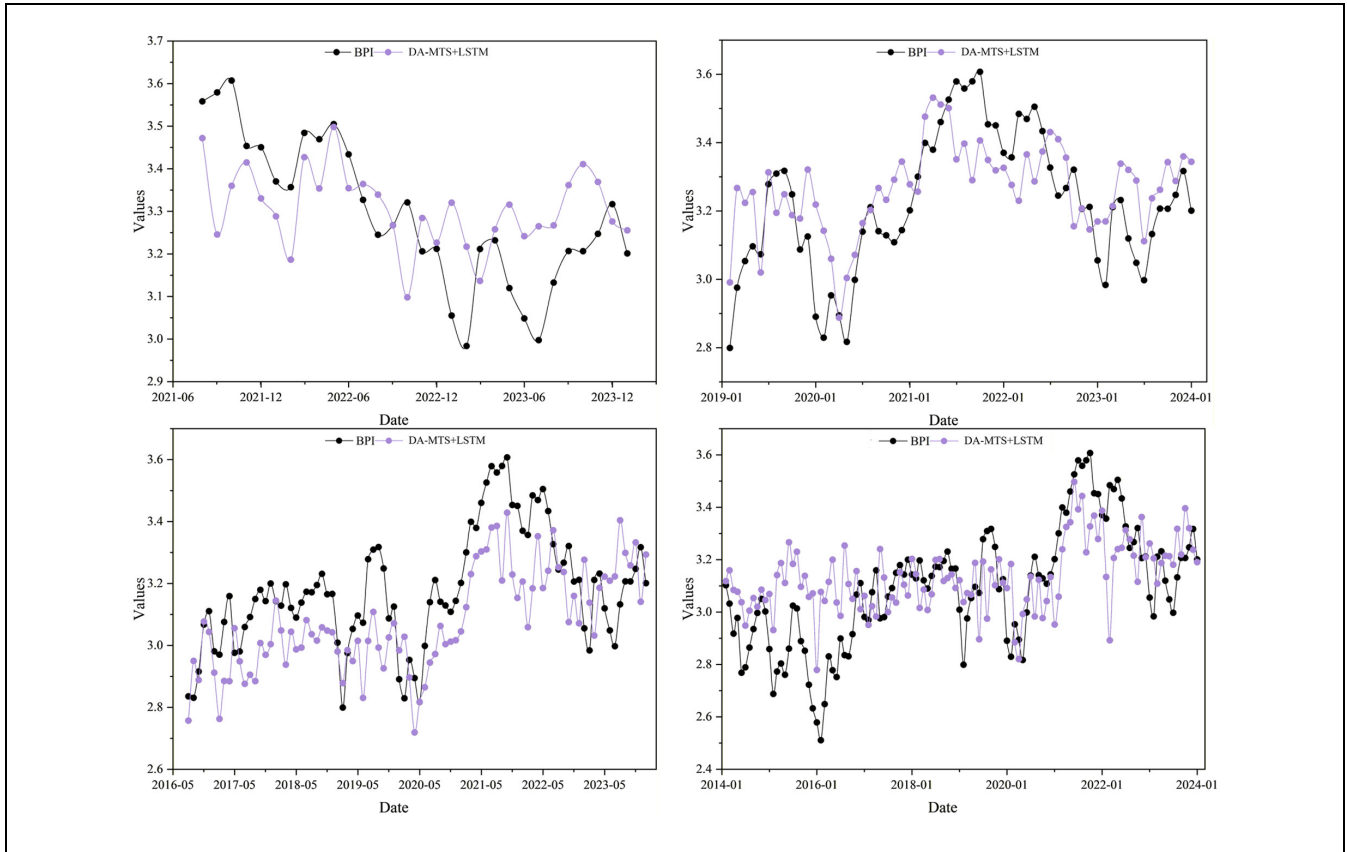


Figure 14. The prediction results of the hybrid model proposed in this paper (DA-MTS+LSTM) for Baltic Panamax Index (BPI) under four different partitioning forms, respectively.

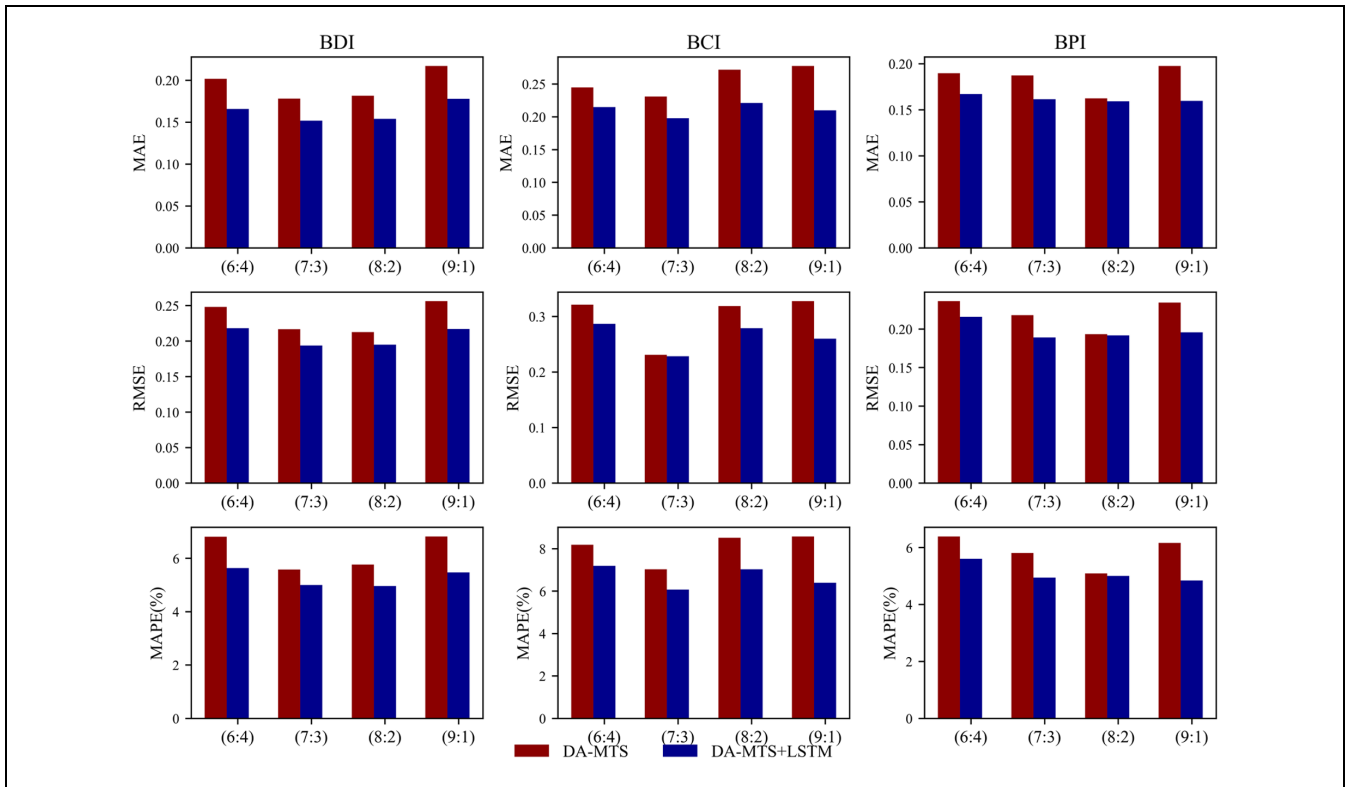


Figure 15. The comparison of prediction errors between the hybrid model proposed in this paper (DA-MTS+LSTM) and domain-adapted maritime time series prediction model (DA-MTS).

Note: MAE, Mean Absolute Error; RMSE, Root Mean Square Error; MAPE, Mean Absolute Percentage Error.

Table 19. Error Evaluation Metrics for Baltic Dry Index(BDI) Forecasting Using the Hybrid Model Proposed in This Paper (DA-MTS+LSTM)

BDI	Category	MAE	RMSE	MAPE, %
(6:4)	DA-MTS	0.2017	0.2482	6.8076
	DA-MTS+LSTM	0.1657	0.2181	5.6330
(7:3)	DA-MTS	0.1715	0.2075	5.1532
	DA-MTS+LSTM	0.1518	0.1938	4.9976
(8:2)	DA-MTS	0.1816	0.2125	5.7621
	DA-MTS+LSTM	0.1539	0.1949	4.9561
(9:1)	DA-MTS	0.1824	0.2316	6.2413
	DA-MTS+LSTM	0.1778	0.2171	5.4715

Note: DA-MTS, domain-adapted maritime time series prediction model; MAE, Mean Absolute Error; RMSE, Root Mean Square Error; MAPE, Mean Absolute Percentage Error.

Table 20. Error Evaluation Metrics for Baltic Capesize Index(BCI) Forecasting Using the Hybrid Model Proposed in This Paper (DA-MTS+LSTM)

BCI	Category	MAE	RMSE	MAPE, %
(6:4)	DA-MTS	0.2448	0.3209	8.1857
	DA-MTS+LSTM	0.2149	0.2866	7.1861
(7:3)	DA-MTS	0.2308	0.2308	7.0256
	DA-MTS+LSTM	0.1977	0.2282	6.0702
(8:2)	DA-MTS	0.2719	0.3186	8.5153
	DA-MTS+LSTM	0.2209	0.2786	7.0251
(9:1)	DA-MTS	0.2775	0.3273	8.5800
	DA-MTS+LSTM	0.2097	0.2596	6.3900

Note: DA-MTS, domain-adapted maritime time series prediction model; MAE, Mean Absolute Error; RMSE, Root Mean Square Error; MAPE, Mean Absolute Percentage Error.

Table 21. Error Evaluation Metrics for Baltic Panamax Index(BPI) Forecasting Using the Hybrid Model Proposed in This Paper (DA-MTS+LSTM)

BPI	Category	MAE	RMSE	MAPE, %
(6:4)	DA-MTS	0.1897	0.2361	6.3882
	DA-MTS+LSTM	0.1670	0.2159	5.6000
(7:3)	DA-MTS	0.1872	0.2178	5.8070
	DA-MTS+LSTM	0.1613	0.1890	4.9356
(8:2)	DA-MTS	0.1623	0.1932	5.0913
	DA-MTS+LSTM	0.1591	0.1916	5.0024
(9:1)	DA-MTS	0.1975	0.2342	6.1600
	DA-MTS+LSTM	0.1596	0.1958	4.8400

Note: DA-MTS, domain-adapted maritime time series prediction model; MAE, Mean Absolute Error; RMSE, Root Mean Square Error; MAPE, Mean Absolute Percentage Error.

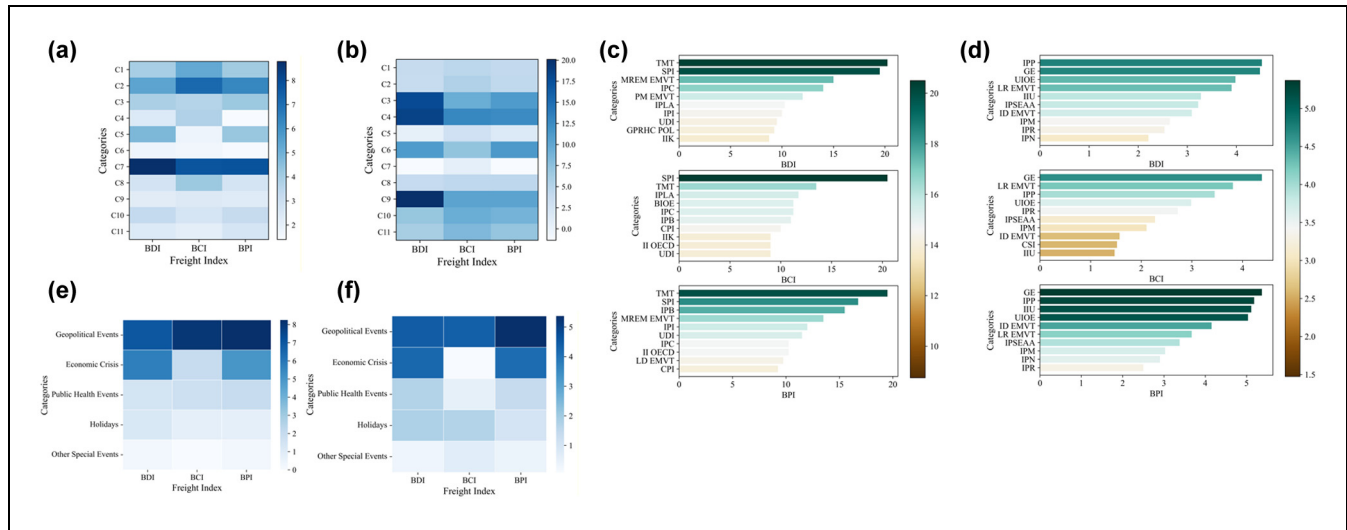


Figure 16. Analysis of factors affecting freight rates: (a) Importance of macro factors on long-term freight rate impact, (b) importance of macro factors on short-term freight rate impact, (c) importance of micro factors on long-term freight rate impact, (d) importance of micro factors on short-term freight rate impact, (e) importance of sudden events on long-term freight rate impact, and (f) importance of sudden events on short-term freight rate impact.

Note: The corresponding indicators in panels (a) and (b) are listed in Table 8, while the abbreviations used in panels (c) and (d) are given in Table 10.

Table 22. Comparison of the Importance of Long-term and Short-term Macroeconomic Factors for Baltic Dry Index (BDI)

Feature name (long-term)	Weight	Feature name (short-term)	Weight (MAE×1000)
Shipping trade indicators	8.79	Geopolitical Risk (GPR) indicators	20.09
Global economic indicators	5.41	Energy price index	18.49
Public health indicators	4.75	Global industrial production indicators	17.81
Commodity trade indicators	3.96	GEPU indicators	11.14
Global industrial production indicators	3.88	U.S. stock market volatility indicators	7.05
U.S. stock market volatility indicators	3.3	Unexpected events	6.00
GPR indicators	2.21	Steel trade indicators	4.00
Steel trade indicators	2.75	Commodity trade indicators	3.92
Energy price index	2.4	Global economic indicators	3.75
Unexpected events	2.4	Public health indicators	0.41
GEPU indicators	1.75	Shipping trade indicators	−1.35

Note: MAE, Mean Absolute Error; GEPU, Global Economic Policy Uncertainty.

Table 23. Comparison of the Importance of Long-term and Short-term Macroeconomic Factors for Baltic Capesize Index (BCI)

Feature name (long-term)	Weight	Feature name (short-term)	Weight (MAE×1000)
Shipping trade indicators	7.75	Energy price index	12.93
Global economic indicators	7.17	Geopolitical Risk (GPR) indicators	10.31
Public health indicators	1.75	Global industrial production indicators	9.44
Commodity trade indicators	5.25	U.S. stock market volatility indicators	9.41
Global industrial production indicators	3.65	Unexpected events	8.31
U.S. stock market volatility indicators	2.64	GEPU indicators	7.25
GPR indicators	2.33	Global economic indicators	5.47
Steel Trade indicators	4.2	Commodity trade indicators	4.65
Energy price index	3.75	Steel trade indicators	4.42
Unexpected events	2.15	Public health indicators	2.94
GEPU indicators	1.64	Shipping trade indicators	0.62

Note: MAE, Mean Absolute Error; GEPU, Global Economic Policy Uncertainty.

Table 24. Comparison of the Importance of Long-term and Short-term Macroeconomic Factors for Baltic Panamax Index (BPI)

Feature name (long-term)	Weight	Feature name (short-term)	Weight (MAE×1000)
Shipping trade indicators	7.79	Energy price index	12.55
Global economic indicators	6.33	GEPU indicators	11.33
Public health indicators	4.25	Global industrial production indicators	11.14
Commodity trade indicators	4.08	Geopolitical Risk (GPR) indicators	10.34
Global industrial production indicators	4.19	U.S. stock market volatility indicators	8.85
U.S. stock market volatility indicators	3.19	Unexpected events	7.17
GPR indicators	2.28	Steel trade indicators	4.54
Steel trade indicators	2.7	Global economic indicators	4.33
Energy price index	1.4	Commodity trade indicators	4.23
Unexpected events	2.65	Public health indicators	1.57
GEPU indicators	1.45	Shipping trade indicators	−1.30

Note: MAE, Mean Absolute Error; GEPU, Global Economic Policy Uncertainty.

Table 25. Comparison of the Importance of Long-term and Short-term Microeconomic Factors for BDI.

Feature name (long-term)	Weight	Feature name (short-term)	Weight (MAE×1000)
TOPIX marine transportation	20.25	Industrial production Poland	4.52
Secondhand price index	19.5	Geopolitical events	4.48
Macro real estate markets equity market volatility tracker	15	Ukraine iron ore exports	3.98
Industrial production China	14	Labor regulations equity market volatility tracker	3.90
Petroleum markets equity market volatility tracker	12	Inflation indicator USA	3.28
Industrial production Latin America	10.25	Industrial production Southeast Asia average	3.22
Industrial production India	10	Public health events	3.09
USA dollar index	9.5	Industrial production Mexico	2.64
Geopolitical risk historical index Poland	9.25	Industrial production Russia	2.53
Inflation indicator South Korea	8.75	Industrial production Netherlands	2.21

Note: BDI = baltic dry index; MAE = mean absolute error.

Table 26. Comparison of the Importance of Long-term and Short-term Microeconomic Factors for BCI

Feature name (long-term)	Weight	Feature name (short-term)	Weight (MAE×1000)
Secondhand price index	20.5	Geopolitical events	4.38
TOPIX marine transportation	13.5	Labor regulations equity market volatility tracker	3.81
Industrial production Latin America	11.75	Industrial production Poland	3.45
Industrial production China	11.25	Ukraine iron ore exports	2.99
Brazil iron ore exports	11.25	Industrial production Russia	2.72
Industrial production Brazil	11	Industrial production Southeast Asia average	2.27
Cotton price index	10	Industrial production Mexico	2.11
USA dollar index	9	Public health events	1.57
Inflation indicator OECD	9	China soybean Imports	1.52
Inflation indicator South Korea	9	Inflation indicator USA	1.47

Note: BCI = baltic capesize index; MAE = mean absolute error.

Table 27. Comparison of the Importance of Long-term and Short-term Microeconomic Factors for BPI

Feature name (long-term)	Weight	Feature name (short-term)	Weight (MAE×1000)
TOPIX marine transportation	19.5	Geopolitical events	5.37
Secondhand price index	16.75	Industrial production Poland	5.18
Industrial production Brazil	15.5	Inflation indicator USA	5.11
Macro real estate markets equity market volatility tracker	13.5	Ukraine iron ore exports	5.03
Industrial production India	12	Public health events	4.15
USA dollar index	11.5	Labor regulations equity market volatility tracker	3.67
Inflation indicator OECD	10.25	Industrial production Southeast Asia average	3.38
Industrial production China	10.25	Industrial production Mexico	3.03
Labor disputes equity market volatility tracker	9.75	Industrial production Netherlands	2.90
Cotton price index	9.25	Industrial production Russia	2.50

Note: BPI = baltic panamax index; MAE = mean absolute error.

Table 28. Comparison of the Importance of Long-term and Short-term Unexpected Events

Categories	Long-term			Short-term (MAE×1000)		
	BDI	BCI	BPI	BDI	BCI	BPI
Geopolitical events	7	8	8.25	4.48	4.38	5.37
Economic crisis	5.75	2	5	4.25	0.09	4.15
Public health events	1.5	1.75	2	1.72	0.56	1.39
Holidays	1.25	0.75	0.75	1.80	1.72	1.02
Other special events	0.25	0	0.25	0.33	0.67	0.40

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; MAE = mean absolute error.

Table 29. A Comparison of the Predictive Performance between the Existing Models and the DA-MTS+LSTM Hybrid Approach.

Categories	Models	BDI			BCI			BPI		
		MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %
SM	ARIMA	0.24	0.29	7.69	0.29	0.35	9.05	0.25	0.30	7.81
	SARIMA	0.27	0.32	8.55	0.34	0.41	10.47	0.28	0.33	8.59
ML	LSTM	0.23	0.27	7.67	0.28	0.35	8.59	0.22	0.24	8.35
	GRU	0.22	0.27	8.75	0.28	0.35	9.52	0.20	0.24	8.37
	RVM	0.22	0.27	6.90	0.29	0.35	8.90	0.21	0.25	6.6
	XGBoost	0.21	0.26	6.70	0.25	0.32	7.87	0.20	0.24	6.23
	Neural ODEs	0.18	0.21	5.46	0.25	0.32	6.95	0.45	0.47	14.1
	ANN	0.53	0.66	16.77	0.63	0.80	19.61	0.45	0.57	14.3
	CNN	0.20	0.23	8.63	0.28	0.33	10.31	0.17	0.20	7.8
Transformer	SVM	0.17	0.21	5.49	0.22	0.28	6.92	0.17	0.21	5.29
	Transformer	0.25	0.32	7.51	0.29	0.38	8.68	0.23	0.31	7.19
	LogSparse-Transformer	0.28	0.37	8.50	0.31	0.40	9.90	0.26	0.33	8.15
	Autoformer	0.27	0.34	8.26	0.29	0.38	8.56	0.23	0.31	6.97
	Informers	0.31	0.41	9.68	0.34	0.43	9.93	0.24	0.32	7.49
	DA-MTS	0.19	0.23	6.24	0.26	0.30	8.08	0.18	0.22	5.84
	LogTrans	0.28	0.37	8.77	0.32	0.41	9.61	0.26	0.34	7.85
This paper	The hybrid model proposed in this paper (DA-MTS+LSTM)	0.16	0.21	5.26	0.21	0.27	6.67	0.16	0.20	5.12

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; MAE = mean absolute error; RMSE = root mean square error; MAPE = mean absolute percentage error; ARIMA = autoregressive integrated moving average; SARIMA = seasonal autoregressive integrated moving average; LSTM = long short-term memory; GRU = gated recurrent unit; RVM = relevance vector machine; ANN = artificial neural network; CNN = convolutional neural network; SVM = support vector machine.

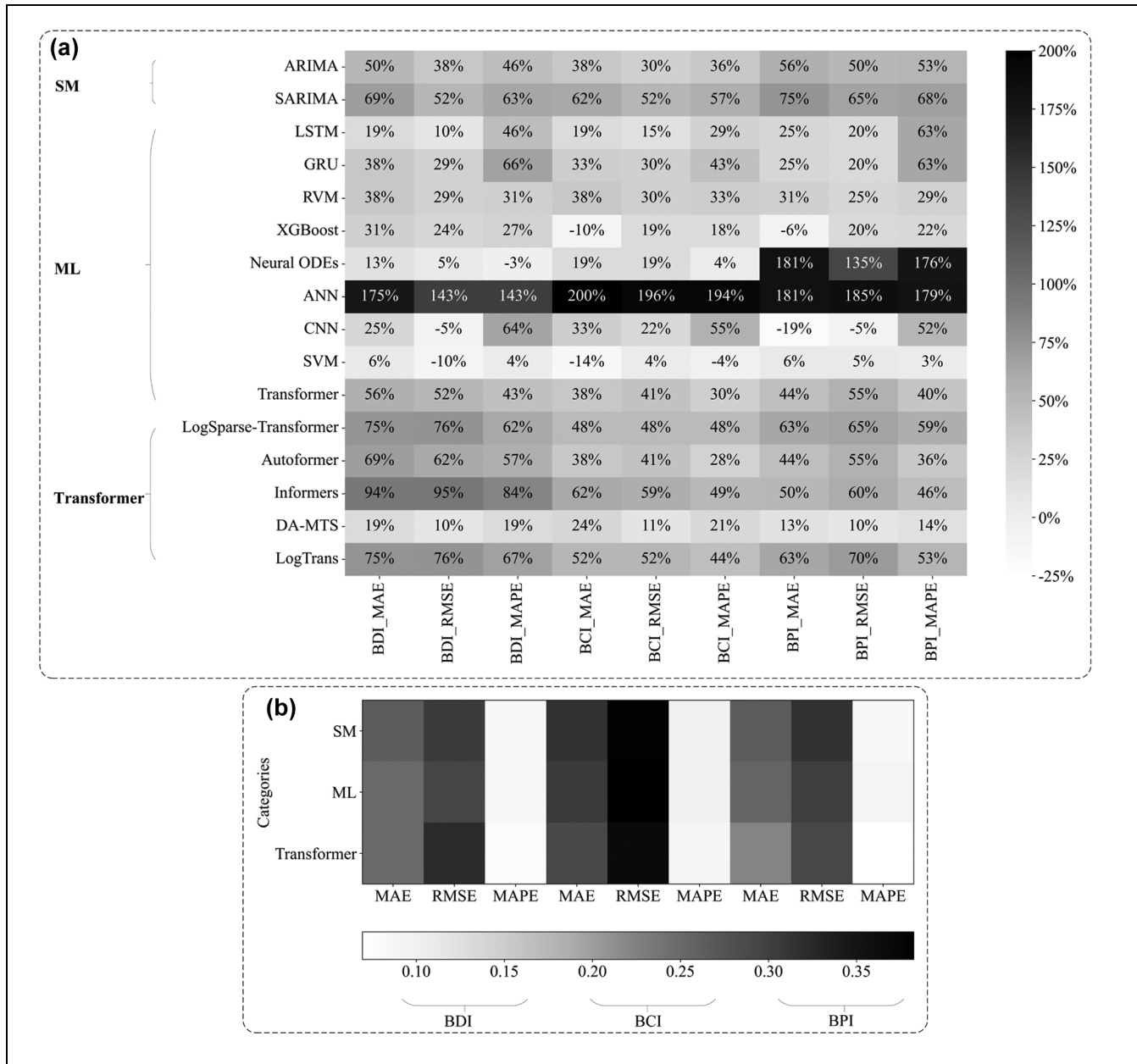


Figure 17. Model performance comparisons: (a) Percentage difference in error of SMs, ML methods, and Transformer-based time series forecast models relative to DA-MTS+LSTM, and (b) comprehensive performance comparison of SM, ML, and DL based on Transformer architecture.

Table 30. Performance Comparison of Different Model Configurations (Ablation Study)

Models	BDI			BCI			BPI		
	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %	MAE	RMSE	MAPE, %
LSTM	0.23	0.27	7.67	0.28	0.35	8.59	0.22	0.24	8.35
DA-MTS	0.19	0.23	6.24	0.26	0.30	8.08	0.18	0.22	5.84
DA-MTS+LSTM	0.16	0.21	5.26	0.21	0.27	6.67	0.16	0.20	5.12

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; MAE = mean absolute error; RMSE = root mean square error; MAPE = mean absolute percentage error; LSTM = long short-term memory; DA-MTS = domain-adapted maritime time series prediction model; DA-MTS-LSTM = the hybrid model proposed in this paper.

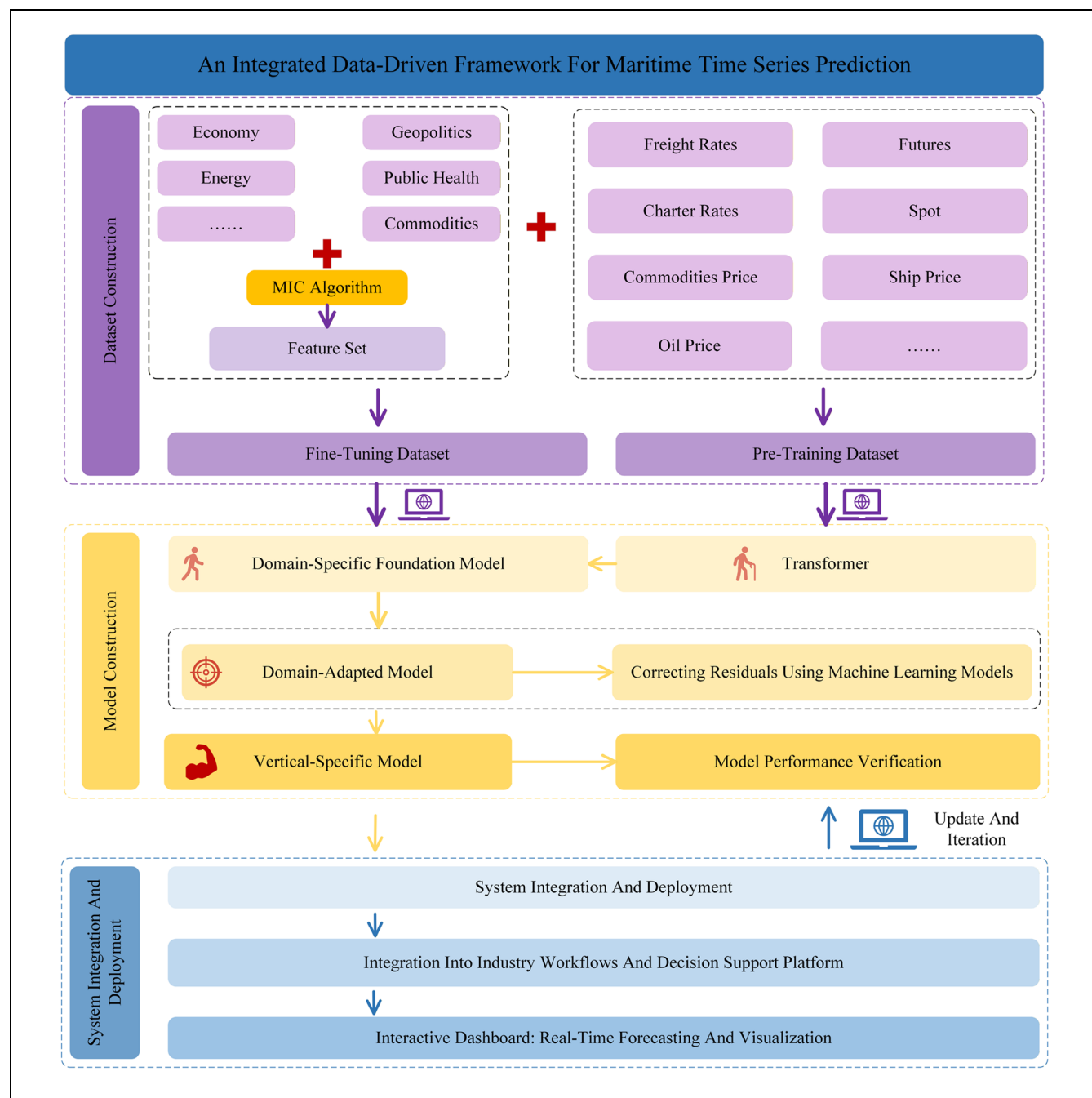


Figure 18. Framework for real-world implementation of the freight transportation prediction model.

Table 31. Comparison of Research Conclusions on Factors Affecting Freight Rates

Variable	Research conclusion	Consistency of conclusion
GPR	GPR has a significant impact on oil prices and freight rates (29). This paper: GPR has a significant short-term impact on freight rates.	Yes
Crude oil prices	Crude oil prices are one of the dominant factors for commodities (such as steel and agricultural products) and freight rates (27). This paper: Crude oil prices have a significant short-term impact on BDI, BCI, and BPI.	Yes
Stock market	The U.S. stock market affects the shipping prices market (51). This paper: Volatility indicators of the U.S. stock market affect freight rates, with a more noticeable impact in the short term.	Yes

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; GPR = geopolitical risk.

Table 32. The Latest Research Findings of this Paper

Category	Research conclusion
Unexpected events	Compared with public health events, large-scale, short-duration, and cyclical holiday events cause more noticeable short-term market sentiment fluctuations, which have a more significant impact on freight rates. Geopolitical events have the most significant short-term impact on freight rates, followed by economic crises, and both have a significantly greater impact on freight rates than public health events. Compared with BPI, BCI is more sensitive to market sentiment fluctuations caused by holidays. Compared with BCI, geopolitical events, economic crises, and public health events have a more significant impact on BPI.
Long-term macro level	Ship trade indicators have the most significant impact on freight rates, followed by global economic indicators.
Short-term macro level	Energy price indices, global industrial production indicators, and GPR indicators have a significant short-term impact on freight rates.
Long-term micro level	The secondhand ship price index, industrial production China, and USA dollar index are the most important for predicting freight rates; BDI, BCI, and BPI are, respectively, sensitive to the macro real estate markets equity market volatility tracker, Brazil iron ore exports, and industrial production Brazil.
Short-term micro level	The labor regulations equity market volatility tracker, Ukraine iron ore exports, industrial production Russian, China soybean imports, and inflation indicator USA have a significantly greater impact on freight rates than other indicators; BDI, BCI, and BPI are, respectively, sensitive to Ukraine iron ore exports, the labor regulations equity market volatility tracker, and the inflation indicator USA.

Note: BDI = baltic dry index; BCI = baltic capesize index; BPI = baltic panamax index; GPR = geopolitical risk.

Bohan Wang, Jibin Zhou; analysis and interpretation of results: Yuping Luo, Rui Zhang, Wenyang Wang, Yajuan Xiang, Bohan Wang, Jibin Zhou; draft manuscript preparation: Yuping Luo. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work is supported by the National Science Fund for Distinguished Young Scholars (T2525020), the National Natural Science Foundation of China (52131203), the Jiangsu Provincial Scientific Research Center of Applied Mathematics (BK20233002), the Humanities and Social Science Fund of the Ministry of Education of China (22YJC910011), the Postdoctoral Research Foundation of China (2023M733444), and the Key R&D Program on Science and Technology Innovation in Artificial Intelligence Field of Liaoning Province of China (2023JH26/10200012).

ORCID iDs

Yuping Luo  <https://orcid.org/0009-0002-8483-395X>
Zhiyuan Liu  <https://orcid.org/0000-0002-6331-0810>

Wenyang Wang  <https://orcid.org/0000-0002-4863-2045>
Bohan Wang  <https://orcid.org/0009-0009-0948-5220>
Jibin Zhou  <https://orcid.org/0000-0002-9269-119X>

Data Accessibility Statement

The data that support the findings of this study are available from the corresponding author on reasonable request.

References

1. Verschuur, J., E. E. Koks, and J. W. Hall. Ports' Criticality in International Trade and Global Supply-Chains. *Nature Communications*, Vol. 13, No. 1, 2022, p. 4351.
2. Najafi, N., and H. Zolfagharinia. Pricing and Quality Setting Strategy in Maritime Transportation: Considering Empty Repositioning and Demand Uncertainty. *International Journal of Production Economics*, Vol. 240, 2021, p. 108245.
3. Schramm, H. J., and Z. H. Munim. Container Freight Rate Forecasting With Improved Accuracy by Integrating Soft facts From Practitioners. *Research in Transportation Business & Management*, Vol. 41, 2021, p.100662.
4. Tang, J., A. Zuo, J. Liu, and T. Li. Seasonal Decomposition and Combination Model for Short-Term Forecasting of Subway Ridership. *International Journal of Machine Learning and Cybernetics*, Vol. 13, 2022, pp. 145–162.
5. Şahin, B., S. Gürgeç, B. Ünver, and İ. Altın. Forecasting the Baltic Dry Index by Using An Artificial Neural Network Approach. *Turkish Journal of Electrical Engineering and Computer Sciences*, Vol. 26, No. 3, 2018, pp. 1673–1684.

6. Caldara, D., and M. Iacoviello. Measuring Geopolitical Risk. *American Economic Review*, Vol. 112, No. 4, 2022, pp. 1194–1225.
7. Saeed, N., S. Nguyen, K. Cullinane, V. Gekara, and P. Chhetri. Forecasting Container Freight Rates Using the Prophet Forecasting Method. *Transport Policy*, Vol. 133, 2023, pp. 86–107.
8. Wang, W., N. He, M. Chen, and P. Jia. Freight Rate Index Forecasting With Prophet Model Based on Multi-Dimensional Significant Events. *Expert Systems with Applications*, Vol. 14, 2024, p. 123451.
9. Yin, K., H. Guo, and W. Yang. A Novel Real-Time Multi-Step Forecasting System With a Three-Stage Data Preprocessing Strategy for Containerized Freight Market. *Expert Systems with Applications*, Vol. 246, 2024, p. 123141.
10. Zhang, F., M. Wang, W. Zhang, and H. Wang. THATSN: Temporal Hierarchical Aggregation Tree Structure Network for Long-Term Time-Series Forecasting. *Information Sciences*, Vol. 692, 2025, p. 121659.
11. Graham, M., J. Peltomäki, and V. Piljak. Global Economic Activity as an Explicator of Emerging Market Equity Returns. *Research in International Business and Finance*, Vol. 36, 2016, pp. 424–435.
12. Bai, X., J. S. L. Lam, and A. Jakher. Shipping Sentiment and the Dry Bulk Shipping Freight Market: New Evidence from Newspaper Coverage. *Transportation Research Part E: Logistics and Transportation Review*, Vol. 155, 2021, p. 102490.
13. Touvron, H., T. Lavril, G. Izacard, X. Martinet, M. A. Lachaux, T. Lacroix, B. Rozière, et al. Llama: Open and efficient foundation language models. *arXiv Preprint arXiv:2302.13971*, 2023.
14. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Ł. Kaiser, and I. Polosukhin. Attention is All You Need. *Proc., 31st International Conference on Neural Information Processing Systems (NeurIPS 2017)*, Long Beach, California, 4–9 December 2017, pp. 5998–6008.
15. Huang, J., Yue, W., Ng, S., and Tsung, F. Overcoming the Semantic Gap in the Customer-to-Manufacturer (C2M) Platform: A Soft Prompts-Based Approach With Pre-trained Language Models. *International Journal of Production Economics*, Vol. 272, 2024, p. 109248.
16. Zhou, H., S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. *Proc., AAAI Conference on Artificial Intelligence*, Vol. 35, No. 12, 2021, pp. 11106–11115.
17. Yang, Z., A. Mitra, W. Liu, D. Berlowitz, and H. Yu. TransformEHR: Transformer-Based Encoder-Decoder Generative Model to Enhance Prediction of Disease Outcomes Using Electronic Health Records. *Nature Communications*, Vol. 14, No. 1, 2023, p. 7857.
18. Wang, J., J. Liu, H. Wang, M. Zhou, G. Ke, L. Zhang, J. Wu, Z. Gao, and D. Lu. A Comprehensive Transformer-Based Approach for High-Accuracy Gas Adsorption Predictions in Metal-Organic Frameworks. *Nature Communications*, Vol. 15, No. 1, 2024, p. 1904.
19. Wang, W., C. Feng, R. Han, Z. Wang, L. Ye, Z. Du, H. Wei, F. Zhang, Z. Peng, and J. Yang. trRosettaRNA: Automated Prediction of RNA 3D Structure With Transformer Network. *Nature Communications*, Vol. 14, No. 1, 2023, p. 7266.
20. Rong, K., Z. Kang, and Y. Luo. Large Models Empower All Industries: An Ecological Business Model. *Chinese Social Sciences Evaluation*, No. 4, 2023, pp. 38–46 + 156.
21. Katris, C., and M. G. Kavussanos. Time Series Forecasting Methods for the Baltic Dry Index. *Journal of Forecasting*, Vol. 40, No. 8, 2021, pp. 1540–1565.
22. Zeng, J., S. Guo, J. Zeng, and K. Jin. Deep Learning Based Method for Early Warning of Price Risk in Railroad Bulk Freight Transport with Consideration of Time Series of Comprehensive Impact Index. *Transportation Research Record: Journal of the Transportation Research Board*, 2024. 2678: 434–449.
23. Zhu, J., Y. Yang, Y. Jin, and W. Liu. Improved Link Prediction Method for Maritime Silk Road Shipping Network Using Composite Index. *Transportation Research Record: Journal of the Transportation Research Board*, 2024. 2678: 674–688.
24. Yang, Z., and E. E. Mehmed. Artificial Neural Networks in Freight Rate Forecasting. *Maritime Economics & Logistics*, Vol. 21, 2019, pp. 390–414.
25. Mo, J., R. Gao, J. Liu, L. Du, and K. F. Yuen. Annual Dilated Convolutional LSTM Network for Time Charter Rate Forecasting. *Applied Soft Computing*, Vol. 126, 2022, p. 109259.
26. Ruan, Q., Y. Wang, X. Lu, and J. Qin. Cross-Correlations Between Baltic Dry Index and Crude Oil Prices. *Physica A: Statistical Mechanics and its Applications*, Vol. 453, 2016, pp. 278–289.
27. Angelopoulos, J., S. Sahoo, and I. D. Visvikis. Commodity and Transportation Economic Market Interactions Revisited: New Evidence from a Dynamic Factor Model. *Transportation Research Part E: Logistics and Transportation Review*, Vol. 133, 2020, p. 101836.
28. Suárez-de Vivero, J. L., and J. C. Mateos. Forecasting Geopolitical Risks: Oceans as Source of Instability. *Marine Policy*, Vol. 75, 2017, pp. 19–28.
29. Monge, M., M. F. R. Rojo, and L. A. Gil-Alana. The Impact of Geopolitical Risk on the Behavior of Oil Prices and Freight Rates. *Energy*, Vol. 269, 2023, p. 126779.
30. Adland, R., P. Cariou, and F. C. Wolff. The Influence of Charterers and Owners on Bulk Shipping Freight Rates. *Transportation Research Part E: Logistics and Transportation Review*, Vol. 86, 2016, pp. 69–82.
31. Xu, L., S. Yang, J. Chen, and J. Shi. The Effect of COVID-19 Pandemic on Port Performance: Evidence from China. *Ocean & Coastal Management*, Vol. 209, 2021, p. 105660.
32. Povey, D., H. Hadian, P. Ghahremani, K. Li, and S. Khudanpur. A Time-Restricted Self-Attention Layer for ASR. *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2018, pp. 5874–5878.
33. Parmar, N., A. Vaswani, J. Uszkoreit, Ł. Kaiser, N. Shazeer, A. Ku, and D. Tran. Image transformer. *International*

- Conference on Machine Learning*. PMLR, 2018, pp. 4055–4064.
34. Jin, K., J. Wi, E. Lee, S. Kang, S. Kim, and Y. Kim. TrafficBERT: Pre-Trained Model With Large-Scale Data for Long-Range Traffic Flow Forecasting. *Expert Systems with Applications*, Vol. 186, 2021, p. 115738.
 35. Xu, C., Y. Wang, and A. Barati Farimani. TransPolymer: A Transformer-Based Language Model for Polymer Property Predictions. *NPJ Computational Materials*, Vol. 9, No. 1, 2023, p. 64.
 36. Zhou, L., and R. H. Zhang. A Self-Attention–Bneural Network for Three-Dimensional Multivariate Modeling and Its Skillful ENSO Predictions. *Science Advances*, Vol. 9, No. 10, 2023, p. eadf2827.
 37. Lai, Z., T. Wu, X. Fei, and Q. Ling. BERT4ST: Fine-Tuning Pre-Trained Large Language Model for Wind Power Forecasting. *Energy Conversion and Management*, Vol. 307, 2024, p. 118331.
 38. Yu, B., H. Yin, and Z. Zhu. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *arXiv Preprint arXiv:1709.04875*, 2017.
 39. Wu, Z., S. Pan, G. Long, J. Jiang, and C. Zhang. Graph Wavenet for Deep Spatial-Temporal Graph Modeling. *arXiv Preprint arXiv:1906.00121*, 2019.
 40. Wang, W., Y. Luo, J. Yu, et al. Analysis and Prediction of China's Methanol Prices Based on Intelligent Chemical Industry Large Models. *Chemical Industry and Engineering Progress*, Vol. 44, No. 10, 2025, pp. 5673–5688.
 41. Gustavsson, M., S. Käll, P. Svedberg, J. S. Inda-Diaz, S. Molander, J. Coria, T. Backhaus, and E. Kristiansson. Transformers Enable Accurate Prediction of Acute and Chronic Chemical Toxicity in Aquatic Organisms. *Science Advances*, Vol. 10, No. 10, 2024, p. eadk6669.
 42. Sun, W., L. C. Chang, and F. J. Chang. Deep Dive Into Predictive Excellence: Transformer's Impact on Groundwater Level Prediction. *Journal of Hydrology*, Vol. 636, No. 2, 2024, p. 131250.
 43. Fang, J., L. Yang, X. Wen, H. Yu, W. Li, J. F. Adamowski, and R. Barzegar. Ensemble Learning Using Multivariate Variational Mode Decomposition Based on the Transformer for Multistep-Ahead Streamflow Forecasting. *Journal of Hydrology*, Vol. 636, 2024, p. 131275.
 44. Mishra, A. K., J. Renganathan, and A. Gupta. Volatility Forecasting and Assessing Risk of Financial Markets Using Multi-Transformer Neural Network Based Architecture. *Engineering Applications of Artificial Intelligence*, Vol. 133, 2024, p. 108223.
 45. Zeng, H., H. Zhang, J. Guo, B. Ren, L. Cui, and J. Wu. A Novel Hybrid STL-Transformer-ARIMA Architecture for Aviation Failure Events Prediction. *Reliability Engineering & System Safety*, Vol. 246, 2024, p. 110089.
 46. Huang, S., Y. Liu, H. Cui, F. Zhang, J. Li, X. Zhang, M. Zhang, and C. Zhang. MEAformer: An All-MLP Transformer With Temporal External Attention for Long-Term Time Series Forecasting. *Information Sciences*, Vol. 669, 2024, p. 120605.
 47. Yu, Y., R. Ma, and Z. Ma. Robformer: A Robust Decomposition Transformer for Long-Term Time Series Forecasting. *Pattern Recognition*, Vol. 15, No. 332, 2024, p. 110552.
 48. Wu, H., J. Xu, J. Wang, and M. Long. Autoformer: Decomposition Transformers With Auto-Correlation for Long-Term Series Forecasting. *Advances in Neural Information Processing Systems*, Vol. 34, 2021, pp. 22419–22430.
 49. Zhou, T., Z. Ma, Q. Wen, X. Wang, L. Sun, and R. Jin. Fedformer: Frequency Enhanced Decomposed Transformer for Long-Term Series Forecasting. *International Conference on Machine Learning*. PMLR, 2022, pp. 27268–27286.
 50. Wang, W., Y. Luo, Y. Xu, D. Liu, J. Zhou, and P. Shao. SPPformer: A Transformer-Based Model With a Sparse Attention Mechanism for Comprehensive and Interpretable Ship Price Analysis. *Transportation Research Part E: Logistics and Transportation Review*, Vol. 199, 2025, p. 104136.
 51. Reshef, D. N., Y. A. Reshef, H. K. Finucane, S. R. Grossman, G. McVean, P. J. Turnbaugh, E. S. Lander, M. Mitzenmacher, and P. C. Sabeti. Detecting Novel Associations in Large Data Sets. *Science*, Vol. 334, No. 6062, 2011, pp. 1518–1524.